

Spontaneous magnetization in the quantum Heisenberg ferromagnet

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Abstract

For every dimension $d \geq 3$ and every spin $S \in \{\frac{1}{2}, 1, \frac{3}{2}, \dots\}$, we prove that the nearest-neighbor isotropic quantum Heisenberg ferromagnet has a translation-invariant, spontaneously magnetized equilibrium state at every sufficiently low positive temperature. The same state satisfies the KMS condition for the zero-field dynamics and has magnetization at least $S/4$. This resolves the low-temperature ordering problem in the spontaneous-magnetization formulation.

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1 Introduction

The nearest-neighbor quantum Heisenberg ferromagnet assigns a quantum spin to each lattice site and favors parallel neighboring spins. Its Hamiltonian is invariant under simultaneous rotations of all spins. Spontaneous magnetization asks whether, at zero applied field, an infinite-volume equilibrium state can nevertheless select a direction and have a nonzero one-site spin expectation. The distinction between an individual ordered state and its rotation-invariant average is essential.

1.1 The model and the result

Fix an integer $d \geq 3$ and a spin $S \in \{\frac{1}{2}, 1, \frac{3}{2}, \dots\}$. At each site use $\mathbb{C}^{2S+1}$ with orthonormal basis $(e_m)_{m=-S}^S$ and operators

$$S^z e_m = m e_m, \quad S^+ e_m = \sqrt{S(S+1) - m(m+1)} e_{m+1}, \quad S^+ e_S = 0.$$

Set $S^- = (S^+)^*$, $S^x = (S^+ + S^-)/2$ and $S^y = (S^+ - S^-)/(2i)$, and write S_x^a for the corresponding operator at a site x . For finite $\Lambda \subset \mathbb{Z}^d$, the free-boundary Hamiltonian at coupling one is

$$H_\Lambda = - \sum_{\substack{\{x,y\} \subset \Lambda \\ |x-y|_1=1}} \sum_{a \in \{x,y,z\}} S_x^a S_y^a. \quad (1.1)$$

The outer sum counts each unordered edge once; in its inner sum the letters $a = x, y, z$ designate spin components.

For a finite set $F \subset \mathbb{Z}^d$, let $\mathcal{A}_F = \bigotimes_{x \in F} M_{2S+1}(\mathbb{C})$, with the usual embeddings obtained by adjoining identity factors. The quasi-local algebra $\mathcal{A}$ is the norm completion of their union. The bounded finite-range interaction in (1.1) defines a dynamics τ_t by the norm limit, on local observables, of

$$\tau_t^\Lambda(A) = e^{itH_\Lambda} A e^{-itH_\Lambda}$$

along increasing boxes. This limit is independent of the boxes. We use the resulting norm-continuous orbit $t \mapsto \tau_t(A)$ for every $A \in \mathcal{A}$. The locality estimates underlying this construction

originate in Lieb and Robinson [23]. Existence and independence of the exhausting boxes follow from [30, Theorem 2.2]; its hypotheses are checked when we use this dynamics in Section 8.2.

A state is a positive linear functional ω on $\mathcal{A}$ with $\omega(1) = 1$. For $0 < \beta < \infty$, it is a β -KMS state for τ if, for every $A, B \in \mathcal{A}$, there is a bounded function $F_{A,B}$, continuous on the closed strip $0 \leq \text{Im } z \leq \beta$ and analytic in its interior, such that

$$F_{A,B}(t) = \omega(A\tau_t(B)), \quad F_{A,B}(t + i\beta) = \omega(\tau_t(B)A) \quad (t \in \mathbb{R}). \quad (1.2)$$

This is the algebraic equilibrium condition developed by Haag, Hugenholtz and Winnink [17]. We call such a state an equilibrium state at inverse temperature β . If T_x denotes translation by $x \in \mathbb{Z}^d$, translation invariance means $\omega \circ T_x = \omega$ for every x .

Theorem 1.1. *For every integer $d \geq 3$ and every $S \in \{\frac{1}{2}, 1, \frac{3}{2}, \dots\}$, there is a finite $\beta_0(d, S) > 0$ such that, for every finite $\beta \geq \beta_0(d, S)$, the dynamics associated with (1.1) has a translation-invariant β -KMS state ω satisfying*

$$\omega(S_0^z) \geq S/4.$$

Thus the zero-field model has spontaneously magnetized equilibrium states at every sufficiently low positive temperature, for each allowed spin. This magnetization is a property of the chosen state and need not be retained by a rotation-invariant average.

1.2 Historical context and significance

Heisenberg's exchange model provided a quantum mechanism for ferromagnetism [18]. Bloch's spin-wave picture predicted the low-temperature reduction of magnetization, and Dyson developed the theory of interacting spin waves and their thermodynamic corrections [3, 12, 13]. These predictions explain why an ordered phase is expected; proving that a zero-field equilibrium state retains a nonzero spin expectation requires control at all length scales at one fixed positive temperature.

The dimension restriction has a rigorous basis. The Mermin–Wagner theorem excludes positive-temperature ferromagnetism in one and two dimensions for finite-range isotropic Heisenberg models [27]. In dimensions at least three, Fröhlich, Simon and Spencer established low-temperature ordering for classical continuous-spin models through infrared bounds [15, Theorem 3.1]. Dyson, Lieb and Simon proved quantum antiferromagnetic ordering for spin at least one in those dimensions, and for spin one half in sufficiently large dimension [14, Theorem 6.2]. Kennedy, Lieb and Shastry subsequently proved ground-state Néel order for the spin-one-half antiferromagnet on the three-dimensional cubic lattice [19]. The corresponding ferromagnetic problem resisted this approach: the reflection-positivity argument does not provide the needed ferromagnetic Duhamel infrared bound, as explained by Dyson, Lieb and Simon [14, pp. 339–340].

Probabilistic representations offer another route. Powers related the isotropic Heisenberg model to random walks on the permutation group [32]. Conlon and Solovej developed random-walk representations for the Heisenberg model and proved a low-temperature free-energy upper bound for spin one half [5, 6]. Tóth's exchange-cycle representation sharpened the pressure bound [36]. General-spin geometric representations were developed by Nachtergaele [29, 28]; the symmetric-slot and endpoint-permutation formulation is also given by Björnberg, Fröhlich and Ueltschi [2, Section 3.2]. Their ordering results concern the complete graph, whereas the representation rests on finite-spin algebra. Section 2 rederives that algebra for the nearest-neighbor interaction and its normalization here. Positive loop weights alone do not establish ordering: the needed additional estimate controls how often an interior cycle avoids pins on the boundary.

Two algebraic methods enter this pin comparison. The stable-polynomial and negative-dependence results of Borcea, Brändén and Liggett [4] underlie the simultaneous pin test in Section 3. The exterior-moment identity used in Section 4 belongs to the determinant-preserving framework of Dereziński, Liang and Mahoney [8]: expected minors agree with the minors of the mean, and this property is preserved under independent matrix products. The conditional next-pin formula needed here is derived in Section 4.

Correggi, Giuliani and Seiringer proved the fixed-spin low-temperature free-energy asymptotics predicted by spin-wave theory in three dimensions [7, Theorem 2.1]. Their bounds also control correlations on a temperature-dependent finite length scale of order $\beta^{5/4}$ [7, equations (2.6)–(2.8)]. More recently, Klippel proved the corresponding free-energy asymptotics in two dimensions [20]. These are thermodynamic and finite-scale advances; the present result constructs an ordered state at a fixed positive temperature in $d \geq 3$.

Lieb explicitly posed the quantum ferromagnetic ordering problem in 1999 [22, Problem A]. Seiringer’s 2025 account records it using a volume-averaged finite-volume two-point order parameter [33]. Theorem 1.1 resolves the low-temperature ordering problem in the spontaneous-magnetization formulation stated here: it constructs a translation-invariant zero-field KMS state selecting a spin direction, for every allowed spin and every $d \geq 3$. The precise conclusion is the one-point state assertion above; the finite-volume correlation criterion in the cited account is a different formulation and is not used in this proof.

1.3 Proof strategy and the main mechanisms

The proof starts in finite cubes. Replace a spin S by $\ell = 2S$ spin- $\frac{1}{2}$ slots at its site and project onto their symmetric subspace. Rate- $\frac{1}{2}$ Poisson transpositions across neighboring sites, together with uniform within-site permutations at the time seam, give an exchange-path representation. Each cycle has weight two, corresponding to its two spin colors. A *pin* is a slot-time point whose cycle is required to have the up color; a cycle meeting a pin has weight one. Section 2 derives the full spectral law of magnetization and a bound on inserted exchange marks under these pin laws.

Pin every boundary slot throughout the time period, and choose a finite time grid. We prove, by downward induction on the number of grid pins, that under every such pin law the probability that m prescribed points all belong to cycles missing the pins is at most p^m , for a small fixed p . The test points lie at one time and form a sparse set: the number in a spatial ball grows at most a small power of its radius. This restriction is enough both to control obstacle components and, at the end, to bound the probability for a single interior slot.

The induction step adds the test set T to the existing pins E , forming $P = E \cup T$. The bound for this larger pin set is already available. Section 3 uses stable multiaffine permutation polynomials to show that the desired avoidance probability under the E -law is at most $(2r)^m$. Here r is the mean fraction of points of T whose lines, under the P -law, return to T at positive elapsed time before reaching E . It therefore suffices to use the larger-pin induction hypothesis to prove $r \leq p/2$.

For this return estimate, expose exactly the cycles that miss P . The unexposed slots support independent heat exchanges and prescribed transports, conditioned on every remaining cycle meeting a pin. Section 4 first identifies this conditional law and its exposure marginal. An exterior-power and determinant calculation then identifies the conditional next-pin first moment with that of a single Markov walk using fresh randomness on successive periods. The exposure itself is still averaged under its original P -law; no independence of the conditioned cycles is required.

Ignore the other interior pins in the walk’s stopping rule, retaining boundary killing. Let U be

its one-period kernel, and let f be one on T and elsewhere the probability of a future visit to T before killing. Section 4 proves

$$m(1 - r) \geq \mathbb{E}_P \langle f, (I - U)f \rangle.$$

The needed lower bound on this quadratic loss is nearly m : a bound of $m(1 - \varepsilon)$ gives $r \leq \varepsilon$. Write $f = \mathbf{1}_T + g$. The two summands are initially disjoint, but their propagated masses can overlap; controlling that overlap is the main analytic task.

The larger-pin induction hypothesis bounds the probability that every site in a prescribed sparse collection has no unexposed slots. Section 5 turns this information into short surviving connections near most anchors and bounds on the sizes of distant obstacle components. The first property supplies a local Nash inequality, which spreads the mass initially on T . The second allows Section 6 to route decaying finite-energy flows through the available sites. These flows control the pairing of the spread mass with the propagated g by its Dirichlet energy at that same time.

Section 7 applies these estimates in short portions at both ends of a period. A weighted time average absorbs the pairing into the energy dissipated by the heat flow, yielding a quadratic loss at least $m(1 - \varepsilon(p))$, with $\varepsilon(p) = o(p)$. Choose p small enough that $2\varepsilon(p) \leq p$; the pin test then closes the finite downward induction. Its singleton conclusion says that under continuous boundary pinning an interior slot is unlikely to lie on a cycle avoiding the boundary.

Finally, requiring every boundary-touching cycle to be up costs only an exponential in the boundary size. The pin estimate therefore gives a macroscopic magnetization tail in the free zero-field Gibbs state. Section 8 selects this tail with a homogeneous field tending to zero as the boxes grow, and averages translations. It verifies both KMS strip boundaries for the same limiting state and the original zero-field dynamics, completing Theorem 1.1.

1.4 A hard-core boson consequence

The spin-one-half correspondence of Matsubara and Matsuda [25, Sections 2–3] turns the ordered state into a state with averaged off-diagonal long-range order. For $S = \frac{1}{2}$, put $b_x = S_x^-$ and $n_x = b_x^\dagger b_x = S_x^z + \frac{1}{2}$. If E_Λ is the set of unordered nearest-neighbor edges in Λ , then (1.1) becomes

$$H_\Lambda = -\frac{1}{2} \sum_{\{x,y\} \in E_\Lambda} (b_x^\dagger b_y + b_y^\dagger b_x) - \sum_{\{x,y\} \in E_\Lambda} (n_x - \frac{1}{2})(n_y - \frac{1}{2}).$$

This is the particle–hole-symmetric attractive hard-core hopping model. For each inverse temperature in Theorem 1.1, rotate the full mean-spin vector $(\omega(S_0^x), \omega(S_0^y), \omega(S_0^z))$ to the positive x direction. Simultaneous spin rotation commutes with the zero-field dynamics and translations, so the rotated state ν is again a translation-invariant β -KMS state. It satisfies $\nu(b_x) = r \geq 1/8$ and $\nu(n_x) = 1/2$, where r is the length of the original mean-spin vector. In particular, it breaks the $U(1)$ symmetry $b_x \mapsto e^{-i\theta} b_x$.

For every nonempty finite Λ , set $B_\Lambda = \sum_{x \in \Lambda} b_x$ and $a_\Lambda = |\Lambda|^{-1/2} B_\Lambda$. Positivity gives $\nu(B_\Lambda^\dagger B_\Lambda) \geq |\nu(B_\Lambda)|^2 \geq |\Lambda|^2/64$, hence

$$\frac{1}{|\Lambda|^2} \sum_{x,y \in \Lambda} \nu(b_x^\dagger b_y) \geq \frac{1}{64}, \quad \nu(a_\Lambda^\dagger a_\Lambda) \geq \frac{|\Lambda|}{64}.$$

The first bound gives averaged off-diagonal long-range order; the diagonal contribution is only $1/(2|\Lambda|)$. Since $\nu(\sum_{x \in \Lambda} n_x) = |\Lambda|/2$, the uniform-mode occupation is at least $1/32$ of the expected particle number. These bounds concern restrictions of the chosen infinite-volume state, not finite-volume Gibbs states.

1.5 Conventions

Unless a different norm is displayed, spatial distances and balls use the sup norm on $\mathbb{Z}^d$; lattice edges always join nearest neighbors. All matrix kernels for one-particle motion act on column vectors of masses, so their columns describe a fixed starting point. Inner products and ℓ^p norms on slots use counting measure. Unless explicitly indicated otherwise, constants may depend on d and S and on explicitly fixed geometric parameters, but not on the volume, the pin set, or the time period. The order of choosing the small pinning parameter and the geometric scales is specified in Sections 5 and 7.

2 Spin slots, loops, and pins

We first express the finite-volume Gibbs state by exchange paths. The representation will identify the entire spectral distribution of total magnetization, which is needed when a small field is introduced at the end of the proof. We then define the pin laws used in the intervening probabilistic argument and establish their two elementary measure bounds. Powers related the isotropic Heisenberg model to random walks on permutations [32]. Tóth gave the random-stirring representation for spin 1/2 [36]. Nachtergaele developed general-spin symmetric-slot representations [28, Section 2.2, equations (2.23)–(2.25), and Theorem 2.4]; see also his account of geometric representations [29]. For the endpoint-permutation realization and the magnetization transform, see also Björnberg, Fröhlich and Ueltschi [2, Section 3.2, Proposition 3.2 and equation (3.19)]. Their representation is presented for the complete graph with a different coupling normalization. We derive the finite-graph algebra here for our nearest-neighbor Hamiltonian, including its rates and full spectral law, before introducing the pin weights needed in the proof.

Fix a finite cube $\Lambda \subset \mathbb{Z}^d$, $S \in \{\frac{1}{2}, 1, \frac{3}{2}, \dots\}$, and $0 < \beta < \infty$. Set $\ell = 2S$ and introduce the *slot set*

$$V_\Lambda = \Lambda \times \{1, \dots, \ell\}.$$

A slot $i = (x, a)$ is located at the site x . Two slots form a *slot edge* if their sites are nearest neighbors; all ℓ^2 pairs above each lattice edge are included. Let $\mathcal{E}_\Lambda$ be this set of unordered slot edges and put $q_\Lambda = |\mathcal{E}_\Lambda|$. On

$$\widehat{\mathcal{H}}_\Lambda = \bigotimes_{i \in V_\Lambda} \mathbb{C}^2$$

write s_i^a for the spin- $\frac{1}{2}$ matrix in direction a acting on slot i . For a permutation π of the slots, $\mathcal{U}_\pi$ denotes the tensor permutation sending the factor originally at i to $\pi(i)$; in particular $\mathcal{U}_\pi \mathcal{U}_\rho = \mathcal{U}_{\pi \circ \rho}$.

Lemma 2.1 (Symmetric-slot realization). *At each site x , the operators $\widehat{S}_x^a = \sum_{j=1}^\ell s_{(x,j)}^a$ restrict on $\text{Sym}^\ell(\mathbb{C}^2)$ to the spin- S matrices defined in Section 1.1. The orthogonal projection onto this subspace is*

$$\mathcal{S}_x = \frac{1}{\ell!} \sum_{\kappa \in \mathfrak{S}_\ell} \mathcal{U}_{\kappa, x}.$$

With $\mathcal{S}_\Lambda = \prod_{x \in \Lambda} \mathcal{S}_x$, define

$$\widehat{H}_\Lambda = - \sum_{\{x,y\} \subset \Lambda: |x-y|=1} \sum_{a=x,y,z} \widehat{S}_x^a \widehat{S}_y^a, \quad \widehat{M}_\Lambda = \sum_{i \in V_\Lambda} s_i^z.$$

Both operators commute with every within-site permutation and with S_Λ . Their restrictions to $\text{ran } S_\Lambda$ are unitarily equivalent to H_Λ and $M_\Lambda = \sum_{x \in \Lambda} S_x^z$, respectively. Moreover,

$$\hat{H}_\Lambda = -\frac{1}{2} \sum_{\{i,j\} \in \mathcal{E}_\Lambda} \mathcal{U}_{(ij)} + \frac{q_\Lambda}{4} I. \quad (2.1)$$

Proof. Let ψ_k be the normalized sum of the $\binom{\ell}{k}$ elementary tensor vectors having exactly k up spins, for $0 \leq k \leq \ell$. These form an orthonormal basis of the symmetric subspace, and

$$\hat{S}^z \psi_k = (k - \ell/2) \psi_k.$$

For $0 \leq k < \ell$, each elementary vector with $k+1$ up spins occurs $k+1$ times in $\sum_j s_j^+ \psi_k$. Consequently

$$\hat{S}^+ \psi_k = (k+1) \sqrt{\frac{\binom{\ell}{k+1}}{\binom{\ell}{k}}} \psi_{k+1} = \sqrt{(\ell-k)(k+1)} \psi_{k+1}.$$

Also $\hat{S}^+ \psi_\ell = 0$. Under $m = k - S$, the squared coefficient is $S(S+1) - m(m+1)$. The lowering, x , and y matrices follow by adjunction and the definitions. Thus $e_m \mapsto \psi_{S+m}$ gives the claimed unitary identification.

The average over the finite permutation group is self-adjoint and idempotent, and its range is exactly the invariant subspace, proving the projection formula. Conjugation by a within-site permutation merely reorders the summands of each $\hat{S}_x^a$. It therefore commutes with $\hat{H}_\Lambda$ and $\hat{M}_\Lambda$. Equivalently, in the slot-edge expression, conjugation permutes the complete collection of ℓ^2 slot pairs above every incident lattice edge. This last observation is essential: an individual inter-site transposition need not commute with S_x .

Finally, on two spin- $\frac{1}{2}$ factors, direct calculation in the up/down basis gives

$$\sum_{a=x,y,z} s_i^a s_j^a = \frac{1}{2} \mathcal{U}_{(ij)} - \frac{1}{4} I.$$

Summing this identity over the slot edges proves (2.1) and the asserted restrictions. $\square$

The *base exchange law* $\mathbb{P}_b$ is defined as follows. Independently on each edge of $\mathcal{E}_\Lambda$, put a Poisson process of rate $1/2$ on the time interval $[0, \beta)$. At a mark, exchange the two slot lines. At the end of the interval, apply an independent uniform permutation of the ℓ slots at each site, and then identify time β with time 0. An *exchange picture* ω comprises these marks and endpoint permutations. Following its lines forward through repeated copies of the same picture gives directed loops, which we call its *cycles*. They correspond to the cycles of its one-period slot permutation π_ω and can wind through several periods. Let $c(\omega)$ be their number. In particular, $c(\omega) \leq |V_\Lambda|$.

Slot-time points belong to $V_\Lambda \times (\mathbb{R}/\beta\mathbb{Z})$. At an exchange we use the slot just after the exchange; time 0 is understood just after the within-site endpoint permutations and the wrapping of time. Thus every cycle has a well-defined set of slots at the time-zero cut. Write a_C for the number of these slots on a cycle C ; then $\sum_C a_C = |V_\Lambda|$.

Proposition 2.2 (Exact spin-to-loop representation). *For every finite cube Λ , spin S , and $0 < \beta < \infty$ as above, the partition function is*

$$Z_\Lambda(\beta) := \text{Tr}_{\mathcal{H}_\Lambda} e^{-\beta H_\Lambda} = e^{\beta q_\Lambda/4} \mathbb{E}_b 2^{c(\omega)}. \quad (2.2)$$

More generally, for every function Φ on the finite spectrum of M_Λ ,

$$\frac{\text{Tr}_{\mathcal{H}_\Lambda}(e^{-\beta H_\Lambda} \Phi(M_\Lambda))}{Z_\Lambda(\beta)} = \frac{\mathbb{E}_b \sum_{\varepsilon \in \{-1,1\}^{\{\text{cycles of } \omega\}}} \Phi\left(\frac{1}{2} \sum_C \varepsilon_C a_C\right)}{\mathbb{E}_b 2^{c(\omega)}}. \quad (2.3)$$

Thus the spectral law of M_Λ is obtained by weighting the base picture by $2^{c(\omega)}$, then coloring each cycle independently up or down with equal probabilities, and summing the corresponding $+1/2$ or $-1/2$ over its slots at time 0.

Proof. Let φ_t be the permutation generated by the inter-site marks up to time t , without the endpoint permutations. Independent increments, or the Poisson expansion in time-ordered marks, give

$$\mathbb{E}_b \mathcal{U}_{\varphi_\beta} = \exp \left\{ \frac{\beta}{2} \sum_{\{i,j\} \in \mathcal{E}_\Lambda} (\mathcal{U}_{(ij)} - I) \right\} = e^{-\beta q_\Lambda/4} e^{-\beta \widehat{H}_\Lambda}.$$

Indeed, over an interval of length dt , the expectation is $I + \frac{1}{2} \sum_{\{i,j\}} (\mathcal{U}_{(ij)} - I) dt + o(dt)$; the resulting constant-coefficient matrix differential equation gives the displayed exponential. Averaging the independent endpoint permutations now gives

$$\mathbb{E}_b \mathcal{U}_{\pi_\omega} = e^{-\beta q_\Lambda/4} \mathcal{S}_\Lambda e^{-\beta \widehat{H}_\Lambda}. \quad (2.4)$$

Lemma 2.1 identifies the trace on the range of $\mathcal{S}_\Lambda$ with the physical trace. Moreover, $\widehat{M}_\Lambda$ commutes with every slot permutation, since it is the sum of the same one-slot observable over all slots. Therefore (2.4) yields

$$\text{Tr}_{\mathcal{H}_\Lambda}(e^{-\beta H_\Lambda} \Phi(M_\Lambda)) = e^{\beta q_\Lambda/4} \mathbb{E}_b \text{Tr}_{\widehat{\mathcal{H}}_\Lambda}(\mathcal{U}_{\pi_\omega} \Phi(\widehat{M}_\Lambda)). \quad (2.5)$$

To calculate the last trace, index the elementary up/down tensor basis by $\eta \in \{-1,1\}^{V_\Lambda}$. The diagonal entry of $\mathcal{U}_{\pi_\omega}$ at η is one exactly when η is constant on every cycle of π_ω , and is zero otherwise. For such a basis vector the eigenvalue of $\widehat{M}_\Lambda$ is $\frac{1}{2} \sum_C \varepsilon_C a_C$. Hence

$$\text{Tr}_{\widehat{\mathcal{H}}_\Lambda}(\mathcal{U}_{\pi_\omega} \Phi(\widehat{M}_\Lambda)) = \sum_{\varepsilon \in \{-1,1\}^{\{\text{cycles of } \omega\}}} \Phi\left(\frac{1}{2} \sum_C \varepsilon_C a_C\right).$$

Taking $\Phi = 1$ proves (2.2); dividing (2.5) by it proves (2.3). In particular, taking Φ to be the indicator of any spectral subset proves the claim about the spectral distribution, not only its moments. $\square$

2.1 Pin laws and the boundary limit

The factor two in the representation records the two colors of each cycle. A pin fixes the color of every cycle it meets. This motivates a family of measures in which only the cycles avoiding the pins retain their factor two.

For a deterministic pin set $D \subset V_\Lambda \times (\mathbb{R}/\beta\mathbb{Z})$, let

$$n_D(\omega) = \#\{C : C \text{ is a cycle of } \omega, C \cap D = \emptyset\}, \quad w_D(\omega) = 2^{n_D(\omega)}, \quad Z_D = \mathbb{E}_b w_D.$$

We use finite pin sets and their unions with the continuous boundary set defined below. Their cycle-incidence events are measurable. The *pin law* and its expectation are

$$\mathbb{P}_D(d\omega) = \frac{w_D(\omega)}{Z_D} \mathbb{P}_b(d\omega), \quad \mathbb{E}_D[\cdot]. \quad (2.6)$$

The set of slot-time points on cycles missing D is denoted by

$$\mathcal{B}_D(\omega) = \bigcup_{C: C \cap D = \emptyset} C.$$

In fixed volume, $1 \leq w_D \leq 2^{|V_\Lambda|}$, so Z_D is finite and strictly positive. The empty-pin law $\mathbb{P}_\emptyset$ is exactly the picture marginal in Proposition 2.2.

The *inner boundary* and the continuous boundary pin set are

$$\partial\Lambda = \{x \in \Lambda : \exists y \notin \Lambda, |x - y|_1 = 1\}, \quad D_0 = \partial\Lambda \times \{1, \dots, \ell\} \times (\mathbb{R}/\beta\mathbb{Z}). \quad (2.7)$$

Every slot above a boundary site is therefore pinned at every time. The following fixed-volume approximation allows finite-layer arguments to be applied with this boundary condition.

Lemma 2.3 (Dense boundary pins). *Fix a finite cube Λ , $0 < \beta < \infty$, and a finite deterministic pin set F . Let $(\mathcal{C}_n)_{n \geq 1}$ be increasing finite sets of times, each containing 0, whose union is dense in $\mathbb{R}/\beta\mathbb{Z}$, and put*

$$D_{0,n} = \partial\Lambda \times \{1, \dots, \ell\} \times \mathcal{C}_n, \quad D_n = F \cup D_{0,n}, \quad D = F \cup D_0.$$

For $\mathbb{P}_b$ -almost every picture there is a finite $n_*(\omega)$ such that, for all $n \geq n_*(\omega)$,

$$w_{D_n}(\omega) = w_D(\omega), \quad \mathcal{B}_{D_n}(\omega) = \mathcal{B}_D(\omega).$$

The laws $\mathbb{P}_{D_n}$ converge to $\mathbb{P}_D$ in total variation. More generally, if $\Phi_n \rightarrow \Phi$ almost surely under $\mathbb{P}_b$ and $\sup_n |\Phi_n| \leq A < \infty$, then

$$\mathbb{E}_{D_n} \Phi_n \longrightarrow \mathbb{E}_D \Phi. \quad (2.8)$$

Proof. In one period the number of Poisson marks is finite almost surely. Their times are distinct and avoid the countable collection of deterministic cut times and pin times, almost surely. A line that visits a boundary site occupies that site on a time interval of positive length: inter-site marks do not coincide, and endpoint permutations preserve the site. This remains true for a visit crossing the time seam. A dense collection of cuts detects every such visit, with every slot included at each boundary cut.

There are only finitely many cycles. For a fixed picture, choose one detecting cut for each cycle that meets D_0 and take the largest of their finite indices. After this index every cycle has the same incidence with D_n as with D . This proves both eventual equalities. Also $w_{D_n} \downarrow w_D$ and $w_{D_n} \leq 2^{|V_\Lambda|}$. Dominated convergence gives $Z_{D_n} \rightarrow Z_D$ and

$$\mathbb{E}_b \left| \frac{w_{D_n}}{Z_{D_n}} - \frac{w_D}{Z_D} \right| \longrightarrow 0,$$

which proves total-variation convergence. Applying dominated convergence to $w_{D_n} \Phi_n$ proves (2.8). The dominating constant here depends on the fixed volume; no uniform-in-volume bound on the total loop weight is asserted. $\square$

2.2 Factorial domination for exchange marks

The pin weights alter the distribution of the Poisson marks. The bound needed later is on ordered lists of distinct marks; it does not require independence under a pin law.

Let $X_\Lambda = \mathcal{E}_\Lambda \times [0, \beta)$ and let μ be the measure satisfying $\mu(\{e\} \times dt) = \frac{1}{2} dt$. Write ω_m for the counting measure of inter-site marks in the picture. For $n \geq 1$, the notation $(\omega_m)_{\neq}^n$ means ordered n -tuples of distinct marks, even when some of their edge labels agree.

Lemma 2.4 (Factorial-measure domination). *For every finite pin set D , or such a set united with D_0 , every integer $n \geq 1$, and every nonnegative measurable function $F : X_\Lambda^n \rightarrow [0, \infty]$,*

$$\mathbb{E}_D \sum_{(u_1, \dots, u_n) \in (\omega_m)_{\neq}^n} F(u_1, \dots, u_n) \leq 2^n \int_{X_\Lambda^n} F(u_1, \dots, u_n) \prod_{j=1}^n \mu(du_j). \quad (2.9)$$

In particular, the estimate holds with the marks restricted to any specified time interval and with any specified chronological order.

Proof. Inserting one inter-site transposition into a picture cuts two directed strands and reconnects them with their outgoing portions exchanged. If they lie on different cycles, those cycles join; if they lie on one cycle, that cycle splits into two. The union of their slot-time strands, and hence all pin incidences on that union, is retained. On joining, the number of cycles missing D cannot increase. On splitting, it increases by at most one: a cycle missing D gives two such cycles, while a cycle meeting D has at least one descendant still meeting D . Thus, away from coincident mark times,

$$w_D \left(\omega + \sum_{j=1}^n \delta_{u_j} \right) \leq 2^n w_D(\omega). \quad (2.10)$$

For finite pins coincidences with pin times have product-intensity measure zero; including D_0 does not change the surgery argument. The endpoint permutations are held fixed throughout this insertion.

The identity used to integrate (2.10) is the iterated Mecke formula [26]; its factorial form is stated in [21, Theorem 4.4]. We derive it here under the base Poisson law, with the pin weight retained inside the integrand. Write $\lambda = \mu(X_\Lambda)$. For a fixed endpoint permutation, expand the Poisson measure as

$$e^{-\lambda} \sum_{k=0}^{\infty} \frac{1}{k!} \prod_{j=1}^k \mu(du_j).$$

In the weighted sum over ordered distinct n -tuples, the k -mark term has $k!/(k-n)!$ choices of the distinguished marks. Relabel them $u_1, \dots, u_n$ and sum the remaining $k-n$ marks. Averaging also over the independent endpoint permutations gives the exact identity

$$\begin{aligned} \mathbb{E}_b \left[w_D(\omega) \sum_{(u_1, \dots, u_n) \in (\omega_m)_{\neq}^n} F(u_1, \dots, u_n) \right] \\ = \int_{X_\Lambda^n} F(u_1, \dots, u_n) \mathbb{E}_b \left[w_D \left(\omega + \sum_{j=1}^n \delta_{u_j} \right) \right] \prod_{j=1}^n \mu(du_j). \end{aligned}$$

All terms are nonnegative, so this manipulation is justified also when an integral is infinite. By (2.10), the expectation inside the integral is at most $2^n Z_D$. Division by Z_D proves (2.9). $\square$

The estimate is unconditional under $\mathbb{P}_D$. If an independent fresh Poisson family is also sampled, mixed lists of distinct actual and fresh marks satisfy the corresponding product bound: an actual mark contributes 2μ , and a fresh mark contributes μ . Indeed, fix which entries in the list come from each family, use independence to factor the two factorial measures, and apply Lemma 2.4 to the actual-mark factor. This observation will permit chronological path estimates after averaging over an exchange picture and fresh randomness; it makes no conditional factorial-domination assertion at a fixed path exposure.

We now have the exact spectral representation and a family of pin laws with controlled mark counts. The next section estimates the probability that many prescribed slot-time points simultaneously belong to cycles avoiding the existing pins.

3 A simultaneous pin test

We relate the probability that many specified points lie on cycles missing the pins to a mean return probability after those points are pinned. The estimate uses the complete pin law, before any trajectories are exposed.

Fix a finite volume and a finite period $\beta > 0$. Let E be a finite pin set, or the union of a finite pin set with D_0 , and let T be a nonempty finite set of slot points at one time, disjoint from E . Put $m = |T|$ and $P = E \cup T$. For $i \in T$, let $\mathcal{R}_i(E, T)$ be the event that the line started at i reaches T again, at strictly positive elapsed time, before it reaches E . The same sampled exchange picture is repeated when following this line through successive periods. Define

$$r = r(E, T) := \frac{1}{m} \sum_{i \in T} \mathbb{P}_P(\mathcal{R}_i(E, T)). \quad (3.1)$$

Thus $0 \leq r \leq 1$. The positive elapsed-time convention excludes the starting point from the return test.

Proposition 3.1 (Simultaneous pin test). *For every finite volume, every $\beta > 0$, every pin set E that is finite or a finite set together with D_0 , and every nonempty finite single-time set T disjoint from E , the return fraction in (3.1) satisfies*

$$\mathbb{P}_E(T \subseteq \mathcal{B}_E) \leq (2r)^{|T|}. \quad (3.2)$$

We first assume that E , and hence $P = E \cup T$, is finite. For each exchange picture, let σ be the permutation of P that maps each pin to the next pin on its directed cycle, at strictly positive elapsed time. The random subset $\sigma(T)$ has exactly m elements. Its relation to the event we want to bound is

$$\{T \subseteq \mathcal{B}_E\} = \{\sigma(T) = T\} = \{T \subseteq \sigma(T)\}. \quad (3.3)$$

Indeed, if every cycle meeting T avoids E , its successive pins all belong to T . Conversely, if $\sigma(T) = T$, iterating σ from a point of T visits every pin on that cycle without reaching E . The last equality uses $|\sigma(T)| = |T|$.

The return fraction also has a subset interpretation:

$$mr = \mathbb{E}_P |T \cap \sigma(T)| = \sum_{j \in T} \mathbb{P}_P(j \in \sigma(T)). \quad (3.4)$$

Bijectivity of σ makes the returning starting points and their outputs in T have the same cardinality. We will prove stability of the subset generating polynomial $\mathbb{E}_P \prod_{j \in \sigma(T)} y_j$ and deduce that the

probability of containing all of T is at most the product of its presence probabilities. By (3.4) that product is at most r^m . A final comparison between the laws $\mathbb{P}_E$ and $\mathbb{P}_P$ supplies the factor 2^m in (3.2).

3.1 The stability operations

Write $\mathbb{H} = \{z \in \mathbb{C} : \text{Im } z > 0\}$. A nonzero polynomial in finitely many complex variables is *stable* if it has no zero when every variable belongs to $\mathbb{H}$. A polynomial is *multiaffine* if its degree in each variable separately is at most one.

Lemma 3.2 (Closure operations). *For polynomials in a fixed finite set of variables the following operations preserve stability, with the possibility of producing the zero polynomial: coefficientwise limits with a common degree bound, specialization of a variable to a real number, and partial differentiation. Identifying two variables preserves stability. In particular, identifying two variables as t and extracting the coefficient of t preserves stability up to zero.*

Proof. Let F_n be stable polynomials of bounded degree converging coefficientwise to a nonzero polynomial F . If F vanished at a point of a product of upper half-planes, choose a complex line through that point on which the restriction of F is not identically zero. A sufficiently small disk in this line lies in the product of upper half-planes. The restrictions of F_n converge uniformly on the disk. On a smaller circle avoiding the isolated zeros of the restriction of F , the argument principle (or Rouché's theorem) forces F_n to have a zero inside for large n . This is impossible. The existence of the line follows, for example, by choosing a direction where the first nonzero homogeneous term of the Taylor expansion of F at the purported zero does not vanish.

Specialization at $a \in \mathbb{R}$ is the limit of specialization at $a + i/n$, so the first assertion applies. Identification of variables is immediate from the definition.

For differentiation in t , let k be the degree of F in t . The case $k = 0$ gives the zero polynomial. If $k \geq 1$, the leading coefficient is the nonzero coefficientwise limit

$$\lim_{R \rightarrow \infty} (iR)^{-k} F(z_1, \dots, z_{n-1}, iR).$$

It is therefore stable. Consequently, after fixing all other variables in $\mathbb{H}$, the resulting polynomial in t still has degree k . Write its roots, with multiplicity, as $\zeta_1, \dots, \zeta_k$; each has imaginary part at most zero. For $t \in \mathbb{H}$,

$$\frac{\partial_t F}{F} = \sum_{j=1}^k \frac{1}{t - \zeta_j}$$

has strictly negative imaginary part. It cannot vanish, proving stability of the derivative. Finally, taking the linear coefficient after identification is differentiation in t followed by $t = 0$. $\square$

The next operation is the partial-symmetrization principle in the stable-polynomial approach to negative dependence [4, Theorem 4.20]. For an elementary proof using complex bivariate sections, see also Liggett [24, Section 5]. We give the calculation in the form needed here.

Lemma 3.3 (Partial symmetrization). *Let F be a stable multiaffine polynomial, let τ exchange two of its variables, and let $0 \leq u \leq 1$. Then $(1 - u)F + u\tau F$ is stable and multiaffine.*

Proof. Fix every other variable in $\mathbb{H}$ and write the two-variable section as

$$q(z, w) = a + bz + cw + ezw,$$

where a, b, c, e can be complex. For every $z \in \mathbb{H}$,

$$N(z) := \operatorname{Im}((a + bz)\overline{(c + ez)}) \geq 0. \quad (3.5)$$

Indeed, if $c + ez \neq 0$, the root $-(a + bz)/(c + ez)$ in the w variable cannot belong to $\mathbb{H}$; if $c + ez = 0$, the displayed quantity is zero. The swapped section $q(w, z)$ is also stable, so

$$N^\tau(z) := \operatorname{Im}((a + cz)\overline{(b + ez)}) \geq 0.$$

Put $b_u = (1 - u)b + uc$ and $c_u = (1 - u)c + ub$. Direct expansion gives

$$\operatorname{Im}((a + b_u z)\overline{(c_u + ez)}) = (1 - u)N(z) + uN^\tau(z) + u(1 - u)|b - c|^2 \operatorname{Im} z. \quad (3.6)$$

If $0 < u < 1$ and $b \neq c$, the right-hand side is strictly positive. In particular $c_u + ez \neq 0$, and the root in w of the mixed section lies strictly below the real axis. If $u \in \{0, 1\}$ or $b = c$, the section is the original section or its swap. These cases also exclude zeros in $\mathbb{H}^2$. Since the other variables were arbitrary, the mixed polynomial is stable; multiaffinity is preserved directly. $\square$

The complex coefficients in this proof matter: other variables have already been fixed in the upper half-plane. No reality assumption on those sections was used.

3.2 Layering the exchange picture and removing unpinned points

For a permutation π of a finite set V , define its polynomial

$$F_\pi(\mathbf{x}, \mathbf{y}) := \prod_{i \in V} (x_i + y_{\pi(i)}).$$

Each x_i and each y_i occurs with degree at most one.

Lemma 3.4 (Stable exchange symbols). *For a permutation obtained by a finite product of independent Bernoulli transpositions and deterministic permutations on a finite set V , the polynomial $\mathbb{E}F_\pi$ is stable. Here each Bernoulli transposition uses an arbitrary probability in $[0, 1]$. The same conclusion holds for any limit of these permutation laws.*

Every finite independent product of finite-time Poisson exchange permutations, uniform permutations on specified subsets of V , and deterministic permutations has such a limit representation.

Proof. For the identity permutation the polynomial is $\prod_{i \in V} (x_i + y_i)$, a product of nonvanishing upper-half-plane sums. Composing the output with a transposition either leaves the polynomial unchanged or interchanges the corresponding two y variables. Averaging an independent Bernoulli choice is precisely the operation in Lemma 3.3. Deterministic permutations only relabel variables. Repetition proves the first assertion. Convergence of permutation laws gives coefficientwise convergence of their symbols, whose degrees are bounded and whose value at all variables equal to 1 is $2^{|V|}$. Lemma 3.2 therefore proves stability of every such limit, with zero excluded.

For a finite-time Poisson exchange process, partition its time interval into slices of length at most δ . On each edge and slice record whether the Poisson clock has at least one mark; these indicators are independent Bernoulli variables. Apply the indicated transpositions in a fixed order within a slice. This coupled approximation agrees with the actual process whenever every slice has at most one mark in total. If the total rate is λ and the interval length is t , the probability of disagreement is at most $C\lambda^2(t + \delta)\delta$, which tends to zero. This constructs the required limit of Bernoulli-product laws.

To approximate a uniform permutation on a subset $A \subseteq V$, run Poisson transpositions at positive rates on every pair in A . For $|A| \leq 1$ there is nothing to prove. Otherwise the unit-time transition matrix on the finite group of permutations of A has all entries positive: every target permutation is a finite word in these transpositions, and that word can occur in the prescribed order within unit time with no other marks. The uniform distribution is invariant, since each step is multiplication by a random group element. If the group has M elements and the minimum transition entry is $\eta > 0$, the transition matrix has a common uniform part of mass $M\eta$. Splitting off that part shows that distance from the uniform law contracts by at most $1 - M\eta$ at each unit step (and is already zero if $M\eta = 1$). Hence the law converges to uniform as time tends to infinity. At each finite time the small-slice construction gives a Bernoulli-product approximation. Taking a sequence with both errors tending to zero proves the asserted representation of the uniform law. For a finite independent product, approximate its factors independently; their product laws then converge to the desired product law. $\square$

Now assume $P = E \cup T$ is finite. Cut the time circle at every time represented in P , and include the time 0 cut if necessary. Let V be the disjoint union of a copy of V_Λ at every cut. Assign an endpoint permutation to the interval ending at time 0, so that this cut has the post-completion convention. Following a line from one cut to the next defines a permutation π of V . Its cycles are exactly the cycles of the exchange picture, with the cut points inserted as vertices.

The random interval bijections are independent. Moreover, the x variables on an interval's starting layer and the y variables on its ending layer are disjoint from the variable sets used by all other intervals. Consequently

$$F_V(\mathbf{x}, \mathbf{y}) := \mathbb{E}_b \prod_{i \in V} (x_i + y_{\pi(i)}) \quad (3.7)$$

is the product of the stable interval symbols of Lemma 3.4, with cyclic relabeling of their output variables. It is therefore stable. This also covers a single cut, where there is just one interval wrapping around the circle.

The next-pin permutation σ is obtained from this layered permutation by following its cycles between successive points of P . Let $n_P(\pi)$ denote the number of cycles of π missing P .

Lemma 3.5 (Splicing and output extraction). *Starting from (3.7), eliminate each $q \in V \setminus P$ by setting $x_q = y_q = t$ and extracting the coefficient of t . The resulting nonzero stable polynomial is*

$$F_P(\mathbf{x}, \mathbf{y}) = \mathbb{E}_b \left[2^{n_P(\pi)} \prod_{i \in P} (x_i + y_{\sigma(i)}) \right]. \quad (3.8)$$

If $T \subseteq P$, then the probability generating polynomial of the random subset $\sigma(T)$ under $\mathbb{P}_P$ is stable. More precisely, with $Z_P = \mathbb{E}_b 2^{n_P(\pi)} > 0$, it is

$$G_T(\mathbf{y}) := \mathbb{E}_P \prod_{i \in T} y_{\sigma(i)} = \frac{1}{Z_P} \left(\prod_{i \in P \setminus T} \partial_{x_i} \right) F_P(\mathbf{x}, \mathbf{y}) \Big|_{x_i=0 \ (i \in P)}. \quad (3.9)$$

Proof. Consider first one deterministic permutation. If q is not a fixed point, write $i = \pi^{-1}(q)$ and $j = \pi(q)$; the two factors involving y_q and x_q become

$$(x_i + t)(t + y_j).$$

Their linear coefficient is $x_i + y_j$. Thus deleting q splices the directed edges $i \rightarrow q \rightarrow j$ into $i \rightarrow j$. This includes a two-cycle, for which $i = j$. If q is a fixed point, its factor is $x_q + y_q = 2t$, giving a factor 2. Successive deletion leaves the next-pin permutation on every cycle meeting P . A cycle missing P eventually becomes a fixed point and contributes exactly one factor 2. Linearity gives (3.8); Lemma 3.2 gives stability up to zero. Its nonnegative coefficients and positive value at all surviving variables equal to 1 exclude zero.

In each product on the right of (3.8), differentiation in x_i , $i \in P \setminus T$, selects the x_i term from exactly those factors. Setting *every* remaining x variable to zero, including those indexed by T , selects $y_{\sigma(i)}$ from each remaining factor. Division by Z_P gives (3.9). Derivatives and real specialization preserve stability up to zero; $G_T(\mathbf{1}) = 1$ excludes zero. Since σ is a permutation, the product contains $m = |T|$ distinct y variables, so this is indeed a multiaffine subset generating polynomial with nonnegative coefficients. $\square$

We have now encoded the outputs of T under the correct pin weight. The next lemma turns stability of this particular generating polynomial into the simultaneous return bound. This is the subset-presence consequence of the strongly Rayleigh framework [4, Theorem 4.9]; the short conditional covariance argument below proves exactly what is required.

3.3 Covariance, normalization, and the dense-boundary limit

Lemma 3.6 (Presence probabilities). *Let A be a random subset of a finite set I . Suppose its generating polynomial $G(\mathbf{y}) = \mathbb{E} \prod_{j \in A} y_j$ is stable. For every $J \subseteq I$,*

$$\Pr(J \subseteq A) \leq \prod_{j \in J} \Pr(j \in A). \quad (3.10)$$

The same assertion holds after conditioning on $B \subseteq A$, whenever that event has positive probability, for subsets of $I \setminus B$.

Proof. Write $X_j = \mathbf{1}_{\{j \in A\}}$. Conditioning on $B \subseteq A$ gives, for the indicators outside B , the generating polynomial

$$\frac{\left(\prod_{j \in B} \partial_{y_j} \right) G(\mathbf{y}) \Big|_{y_j=1 \ (j \in B)}}{\Pr(B \subseteq A)}.$$

It is nonzero and stable by Lemma 3.2. Under this conditional law, keep only two variables z, w and specialize all others to 1. The resulting stable polynomial has the form $a + bz + cw + ez w$, where $a, b, c, e \geq 0$ and $a + b + c + e = 1$. Applying (3.5) with real coefficients gives

$$0 \leq \operatorname{Im}((a + bz)\overline{(c + ez)}) = (bc - ae) \operatorname{Im} z.$$

Thus $bc \geq ae$. The covariance of the corresponding presence indicators equals

$$e - (b + e)(c + e) = ae - bc \leq 0.$$

Whenever the second indicator has positive probability of being one, conditioning it to be one therefore cannot increase the probability that the first is one.

Order $J = \{j_1, \dots, j_k\}$. If any prefix-presence event has probability zero, (3.10) is immediate. Otherwise, for each h , remove the conditions $X_{j_{h-1}} = 1, \dots, X_{j_1} = 1$ one at a time. The preceding conditional covariance inequality gives

$$\Pr(X_{j_h} = 1 \mid X_{j_1} = \dots = X_{j_{h-1}} = 1) \leq \Pr(X_{j_h} = 1).$$

Multiplying the conditional probabilities proves (3.10). Applying the same argument to the conditional generating polynomial proves the final assertion. $\square$

Proof of Proposition 3.1. Suppose first that E is finite, and use the layered construction above with $P = E \cup T$. By Lemmas 3.5 and 3.6, followed by the arithmetic–geometric mean inequality and (3.4),

$$\mathbb{P}_P(T \subseteq \sigma(T)) \leq \prod_{j \in T} \mathbb{P}_P(j \in \sigma(T)) \leq r^m. \quad (3.11)$$

If one of the factors is zero this conclusion is interpreted directly; no conditioning on a zero-probability event is needed.

By (3.3), the event on the left is $H = \{T \subseteq \mathcal{B}_E\}$. It remains to compare its probability under the two pin laws.

For clarity, write $w_D = 2^{n_D}$, where n_D is the number of cycles missing D , and $Z_D = \mathbb{E}_b w_D$. Deleting the m pins in T changes the number of unpinned cycles by the number of cycles that meet T but miss E . That number lies between 0 and m . Hence, picture by picture,

$$w_P \leq w_E \leq 2^m w_P, \quad Z_E \geq Z_P > 0.$$

We obtain

$$\mathbb{P}_E(H) = \frac{\mathbb{E}_b(\mathbf{1}_H w_E)}{Z_E} \leq 2^m \frac{Z_P}{Z_E} \mathbb{P}_P(H) \leq 2^m r^m.$$

This proves the finite-pin case with the required normalization.

Finally let $E = F \cup D_0$ with F finite. Choose increasing finite sets of boundary cuts with dense union, and let $D_{0,n}$ contain all boundary slots at those cuts. Put $E_n = F \cup D_{0,n}$ and $P_n = E_n \cup T$. Then E_n and T are disjoint and the finite-pin inequality applies. As in Lemma 2.3, almost every picture has finitely many marks, and every boundary visit contains an interval of positive duration. Thus every boundary-touching cycle is eventually detected, and

$$w_{E_n} \longrightarrow w_E, \quad w_{P_n} \longrightarrow w_P, \quad \mathbf{1}_{\{T \subseteq \mathcal{B}_{E_n}\}} \longrightarrow \mathbf{1}_{\{T \subseteq \mathcal{B}_E\}}.$$

The weights are bounded by $2^{|V_\Lambda|}$ in this fixed volume, and all their normalizers are positive, so the corresponding normalized probabilities converge by dominated convergence.

The same reasoning applies to the return test, with a useful finite horizon. Starting from $i \in T$, follow the uncensored line until its first positive return to T . This time is finite, since the line eventually returns to i on its finite permutation cycle. If the line meets F before that return, every finite-cut test fails. If it first meets the boundary before that return, one of its positive-duration boundary visits before the return contains a cut from the dense union, so all sufficiently fine tests fail. If it meets neither, every test succeeds. Since T is disjoint from D_0 , a boundary visit cannot first occur only at the return point. Consequently, for every $i \in T$,

$$\mathbf{1}_{\mathcal{R}_i(E_n, T)} \longrightarrow \mathbf{1}_{\mathcal{R}_i(E, T)} \quad \text{almost surely.}$$

Bounded convergence with the weights w_{P_n} now gives $r(E_n, T) \rightarrow r(E, T)$. Taking limits in $\mathbb{P}_{E_n}(T \subseteq \mathcal{B}_{E_n}) \leq (2r(E_n, T))^m$ proves (3.2) for continuous boundary pins. Only scalar probabilities and bounded weights are passed to this limit; no polynomial with infinitely many variables is required. $\square$

4 Exposed paths and a one-particle return identity

The simultaneous test in Proposition 3.1 reduces the probability of many pin-free points to an expected fraction of returns. Our objective here is to express that first moment through a Markov walk. We first describe the conditional exchange picture after all pin-free cycles have been exposed. A determinant identity then removes the remaining cycle condition from the first moment. Finally, we relate the return probability to the loss of a quadratic form under one period of the walk.

Throughout this Section, the volume and period are finite. Let $E = D_0 \cup E_f$, where E_f is a finite set of slot-time points. Let T be a nonempty set of m distinct slot points at one time θ , disjoint from E , and put $P = E \cup T$. Under $\mathbb{P}_P$, let

$$r = \frac{1}{m} \sum_{i \in T} \mathbb{P}_P(\text{the next point of } P \text{ after } i \text{ belongs to } T). \quad (4.1)$$

“After” always means at strictly positive elapsed time. In particular, a return to the same slot point after one or more periods counts.

4.1 Conditional exchanges on the unexposed slots

Label every line by its slot at time 0, immediately after the within-site completion. The one-period endpoint permutation sends these labels to their slots at the next time 0. Expose the initial labels on cycles missing P , their trajectories over one period, and their endpoint mappings. Write $\mathcal{H}$ for the sigma-field generated by these data and $\mathbf{a}$ for an exposure. At time t , let $H_t(\mathbf{a})$ be the set of slots not occupied by an exposed label. We call these slots *holes*. All pins are holes at their respective times; in particular every boundary slot is a hole throughout the period.

The histories are finite-jump paths on a finite state space, together with finite endpoint data. Their spaces, and the finite disjoint union over possible exposed initial sets, are standard Borel spaces. Thus the conditional laws below can be taken as regular conditional probabilities. Equalities of such laws are always asserted almost surely for the specified outer measure.

For a fixed exposure $\mathbf{a}$, define a probability law $\mathbb{Q}_{\mathbf{a}}$ on the remaining exchange picture as follows. Independently put rate-1/2 Poisson exchanges on each slot edge during the time intervals when both endpoints are holes. At a jump of an exposed label from a slot to a hole, transport the hole in the opposite direction, into the vacated slot. This is a prescribed bijection between the hole sets just before and just after that jump. A swap of two exposed labels leaves the holes unchanged. At the end of the period, complete the revealed within-site mappings by independent uniform bijections of the remaining slots at each site. The hole sets and the forced transports are fixed by $\mathbf{a}$; the Poisson exchanges and the completions are the randomness in $\mathbb{Q}_{\mathbf{a}}$.

Let G_P be the event that every cycle of this remaining picture meets P , and define

$$g_P(\mathbf{a}) = \mathbb{Q}_{\mathbf{a}}(G_P). \quad (4.2)$$

The set of exposed labels is selected by the condition that their cycles miss P . To compute its law, we first follow a deterministic label set A under the base law. Requiring those labels to close among themselves and avoid P , and every complementary cycle to meet P , then selects exactly one A in each complete picture. The next proposition sums these disjoint possibilities and records both the conditional law and the exposure marginal used in later averages.

Proposition 4.1 (Exposed-path disintegration). *For P as above, the conditional law of the unexposed picture given $\mathcal{H}$ under $\mathbb{P}_P$ is $\mathbb{Q}_{\mathbf{a}}(\cdot \mid G_P)$, and $g_P(\mathbf{a}) > 0$ for almost every exposure under this pin law.*

More precisely, fix a deterministic initial label set $A \subseteq V_\Lambda$, and let μ_A be the distribution under $\mathbb{P}_b$ of the histories and endpoint mappings of these labels. Let F_A mean that their endpoint set is A and their paths avoid P . On F_A , let $c_A(\mathbf{a})$ be the number of cycles in their endpoint permutation. If Z_P is the normalizer defining $\mathbb{P}_P$, then for every bounded measurable function Φ of the exposure and remaining picture,

$$\begin{aligned} \mathbb{E}_P \Phi(\mathbf{a}, \omega_h) \\ = \frac{1}{Z_P} \sum_{A \subseteq V_\Lambda} \int_{F_A} 2^{c_A(\mathbf{a})} \mathbb{E}_{\mathbb{Q}_A} [\mathbf{1}_{G_P} \Phi(\mathbf{a}, \omega_h)] \mu_A(d\mathbf{a}). \end{aligned} \quad (4.3)$$

Consequently, the exposure marginal on the A -branch is

$$\frac{1}{Z_P} \mathbf{1}_{F_A}(\mathbf{a}) 2^{c_A(\mathbf{a})} g_P(\mathbf{a}) \mu_A(d\mathbf{a}). \quad (4.4)$$

The same conclusions hold with P any finite deterministic pin set.

Proof. We first fix A under the base law and verify the conditional construction before imposing any cycle condition. For each slot edge e , take two independent rate-1/2 Poisson clocks $N_e^{(1)}, N_e^{(2)}$, all independent across edges. Accept a mark of $N_e^{(1)}$ exactly when at least one endpoint contains an A -label immediately before it. Accept a mark of $N_e^{(2)}$ exactly when neither endpoint contains an A -label.

This construction has the base exchange law. To see this directly, if there are b slot edges, the superposed clock has rate b , and its next mark has an independent uniform type among the $2b$ clocks. For each edge exactly one of its two types is accepted, whatever the past configuration. Thus, conditionally on the past, the next superposed mark is accepted on any specified edge with probability $1/(2b)$, and is rejected with probability $1/2$. Iterated conditioning shows that these accepted-edge or rejection outcomes have the fixed product distribution and are independent of the superposed waiting times. Poisson splitting therefore yields independent rate-1/2 accepted processes on the edges. If there are no edges the assertion is immediate.

The motions of the A -labels before the endpoint permutations use only the first clock family: second-family marks always exchange two other labels. At each site, condition a uniform endpoint permutation on its images of the A -labels that are there just before the endpoint. Each bijection between the remaining source and target slots has the same number, namely one, of extensions with those prescribed images. The conditional completion is therefore uniform. Completions at different sites remain independent, since the endpoint permutations were independent and each revealed restriction concerns its own site.

It follows that the full specified histories, including endpoint images, are determined by the first clock family and these revealed restrictions, without using the second family or the remaining completions. More explicitly, the pre-completion locations of the A -labels depend only on the first clock family. The endpoint permutations are independent of both clock families, so conditioning on these locations and the selected endpoint images leaves the second clock family independent. Conditional on each revealed source and target restriction, the remaining bijection is uniform over its allowed completions. Conditional on the histories, the second-family clocks are still independent Poisson clocks. Restricting them to the now fixed intervals with two hole endpoints gives the independent exchanges in $\mathbb{Q}_A$. The cross jumps and the remaining endpoint completions give exactly its other prescribed operations. Although conditioning on histories may leave unobserved first-family marks, those marks are rejected whenever they have two hole endpoints and cannot affect the remaining picture. This proves the claimed base conditional kernel on F_A .

On F_A , the labels of A form a union of whole time cycles missing P . Their number is $c_A(\mathbf{a})$, which is determined by the endpoint mapping of A . Requiring in addition that every remaining cycle hit P makes A exactly the initial labels of all the missing cycles. Therefore for each base picture exactly one initial set satisfies both requirements. If n_P is the number of cycles missing P , the pointwise identity is

$$2^{n_P} = \sum_{A \subseteq V_A} \mathbf{1}_{F_A} 2^{c_A} \mathbf{1}_{G_P}, \quad (4.5)$$

where each summand is defined as zero off F_A . Apply the base conditional kernel on each branch of this finite sum. Dividing by $Z_P = \mathbb{E}_b 2^{n_P}$ gives (4.3). Integrating the remaining picture gives (4.4). Exposures with $g_P(\mathbf{a}) = 0$ consequently have zero outer measure; on its positive set, division by $g_P(\mathbf{a})$ gives the asserted conditional law.

For finite pins the same argument applies without change. For pins including D_0 , avoidance of the boundary is a measurable property of the finite-jump histories. Thus the argument applies directly to that case as well. $\square$

Only the independent law $\mathbb{Q}_a$ will enter the linear-algebra calculation below. Conditioning it on G_P may destroy the exterior-moment identity for the original layered permutation. The factor $g_P(\mathbf{a})$ in (4.4) also remains present when those conditional first moments are subsequently averaged under $\mathbb{P}_P$.

4.2 An exterior-power identity and all-hit conditioning

If M is a matrix on a finite set V , write $\bigwedge^k M$ for its action on the k -th exterior power; its entries are the k -row, k -column minors of M , with respect to the chosen ordering of V . The convention is $\bigwedge^0 M = 1$. Permutation matrices are column-forward: $\Pi e_i = e_{\pi(i)}$.

The all-minors identity below is the determinant-preserving property studied by Dereziński, Liang and Mahoney [8, Definition 4, Example 2, and Lemma 4]. Their rank-one example and closure under independent sums and products give the finite-product case below. We include the rank-one argument and the passage to the exchange law needed here.

Lemma 4.2 (Exterior moments before conditioning). *Let Π be a random permutation matrix on a finite set V . Suppose its law is a limit of laws of finite products of independent Bernoulli transpositions and deterministic permutations. Here a Bernoulli transposition equals a fixed transposition with an arbitrary probability $u \in [0, 1]$, and equals the identity otherwise. Then, writing $K = \mathbb{E}\Pi$,*

$$\mathbb{E}[\bigwedge^k \Pi] = \bigwedge^k K \quad (0 \leq k \leq |V|). \quad (4.6)$$

In particular, fix an exposure from Proposition 4.1 and finitely many time cuts. Under $\mathbb{Q}_a$, the permutation on the disjoint union of the hole layers, obtained by following each hole from a cut to the next cyclic cut, satisfies (4.6).

Proof. If T_{ij} transposes two coordinates, then

$$T_{ij} - I = -(e_i - e_j)(e_i - e_j)^*.$$

Every minor of $I + u(T_{ij} - I)$ is affine in u : a term using two columns of the rank-one perturbation has determinant zero. Therefore

$$\bigwedge^k ((1 - u)I + uT_{ij}) = (1 - u) \bigwedge^k I + u \bigwedge^k T_{ij}.$$

This is (4.6) for one Bernoulli transposition. For independent factors A, B satisfying the identity,

$$\mathbb{E} \bigwedge^k (AB) = \mathbb{E} \bigwedge^k A \mathbb{E} \bigwedge^k B = \bigwedge^k (\mathbb{E} A \mathbb{E} B) = \bigwedge^k \mathbb{E}(AB).$$

Deterministic permutations satisfy the identity as well, so induction proves it for finite products. All the matrix entries and minors are bounded continuous functions on the finite permutation space, which proves passage to limits.

We verify the limit representation for the hole picture. Between successive exposed jumps and endpoint times, the hole set is fixed and the available exchanges have constant rates. Their permutation law has the Bernoulli-product approximation proved in Lemma 3.4. The forced transports are deterministic bijections between equally sized hole sets.

The same lemma supplies this approximation for every uniform completion. When its source and target slot sets differ, precede or follow a uniform permutation of one fixed set by a deterministic bijection.

For completeness, the layered permutation has the same product structure. The total number of holes is constant, so identify every cut layer with one fixed set of that size. The independent interval maps act on separate layer blocks; their direct sum, followed by the fixed cyclic shift of the layers, is the layered permutation. Each random operation within a block is a transposition on the full layered set, and each fixed transport is a permutation there. Independence of exchanges on disjoint intervals and of the completions proves the required representation. $\square$

The return test allows a line started in T to reach a different point of T . If Σ is the column-forward matrix of a next-pin permutation on a finite pin set containing T , its number of returns from T to T is $\sum_{i,j \in T} \Sigma_{ji}$, rather than its trace. A diagonal-only first-moment identity would therefore not suffice. The following finite-dimensional statement recovers every entry of the mean by allowing an arbitrary matrix W .

Proposition 4.3 (Conditional next-pin first moment). *Let $V = J \sqcup Q$ be a finite set with $J \neq \emptyset$. Let Π be a random permutation matrix satisfying (4.6), and put $K = \mathbb{E}\Pi$. Let G_J be the event that every cycle of Π meets J . On G_J , let Σ be the column-forward permutation matrix of the next visit to J at positive time along the cycles. Then*

$$\mathbb{P}(G_J) = \det(I - K_{QQ}). \quad (4.7)$$

If this probability is positive, K_{QQ} is transient and

$$\mathbb{E}[\Sigma \mid G_J] = H := K_{JJ} + K_{JQ}(I - K_{QQ})^{-1}K_{QJ}. \quad (4.8)$$

The matrix H is the first positive J -hitting kernel for the Markov chain with transition matrix K , using independent transitions on successive steps. More strongly, for every complex $J \times J$ matrix W ,

$$\mathbb{E}[\det(I - W\Sigma) \mid G_J] = \det(I - WH). \quad (4.9)$$

An empty Q is allowed, with determinant of the empty matrix equal to one and the Schur correction in (4.8) equal to zero.

For time-homogeneous interchange with a single pin, G_J is the event that the sampled permutation consists of one cycle. In that setting, (4.7) is equivalent to the single-cycle formula of Alon and Kozma [1, Theorem 1].

Proof. For a fixed matrix D , the principal-minor expansion and multiplicativity of exterior powers give

$$\begin{aligned}\mathbb{E} \det(I - D\Pi) &= \sum_{k=0}^{|V|} (-1)^k \operatorname{Tr} \left(\bigwedge^k D \mathbb{E} \bigwedge^k \Pi \right) \\ &= \det(I - DK).\end{aligned}$$

Take $D = \operatorname{diag}(W, I_Q)$. If a permutation cycle lies entirely in Q , its indicator vector v , extended by zero on J , satisfies $D\Pi v = v$, even for arbitrary W . Thus the determinant on that picture is zero. On G_J , Π_{QQ} is nilpotent: a sufficiently long path confined to Q would contain a cycle there. Its inverse resolvent is a finite sum, and block elimination gives

$$\begin{aligned}\det(I - D\Pi) &= \det(I - \Pi_{QQ}) \det \left(I - W \left[\Pi_{JJ} + \Pi_{JQ}(I - \Pi_{QQ})^{-1} \Pi_{QJ} \right] \right) \\ &= \det(I - W\Sigma).\end{aligned}$$

Here $\det(I - \Pi_{QQ}) = 1$, and each term of the resolvent follows the permutation through a specified number of intermediate nonpins. Consequently the preceding determinant identity reads

$$\mathbb{E}[\mathbf{1}_{G_J} \det(I - W\Sigma)] = \det \begin{pmatrix} I - WK_{JJ} & -WK_{JQ} \\ -K_{QJ} & I - K_{QQ} \end{pmatrix}. \quad (4.10)$$

At $W = 0$, this proves (4.7) without any invertibility assumption. If its value is positive, $I - K_{QQ}$ is invertible. The Schur complement on the right of (4.10), followed by division by that positive determinant, gives (4.9).

To extract the first moment, use $\det(I - tWA) = 1 - t \operatorname{Tr}(WA) + O(t^2)$. The linear terms in (4.9) yield

$$\operatorname{Tr}(W\mathbb{E}[\Sigma \mid G_J]) = \operatorname{Tr}(WH) \quad \text{for every } W.$$

Taking the matrix units for W gives each entry of (4.8).

Finally, K is nonnegative and doubly stochastic, as the mean of permutation matrices. Its restriction K_{QQ} is substochastic. If a state in Q could not reach J by any positive-probability path, its reachable set would be a nonempty closed subset of Q . The restriction to that set would have column sums one and eigenvalue one, forcing $I - K_{QQ}$ to be singular. Therefore every state in Q has a positive path to J . Since there are finitely many states, some integer n and number $\varepsilon > 0$ make the probability of leaving Q within n steps at least ε , uniformly over starting states in Q . It follows that

$$\|K_{QQ}^{kn}\|_{1 \rightarrow 1} \leq (1 - \varepsilon)^k, \quad (I - K_{QQ})^{-1} = \sum_{j=0}^{\infty} K_{QQ}^j.$$

The first term of H records a direct next-step hit of J , and the term $K_{JQ}K_{QQ}^jK_{QJ}$ records a first hit after $j + 1$ intervening steps in Q . This proves its Markov interpretation and completes the proof. $\square$

We have now identified the conditional first moment at a finite set of cuts. In the hole construction, each application of K in this Markov interpretation resamples the relevant interval transition. It does not follow a single sampled permutation repeatedly. This distinction permits the return calculation to use one-particle heat flow in the fixed exposed environment. Figure 1 summarizes the finite-cut reduction and the two probability laws it distinguishes.

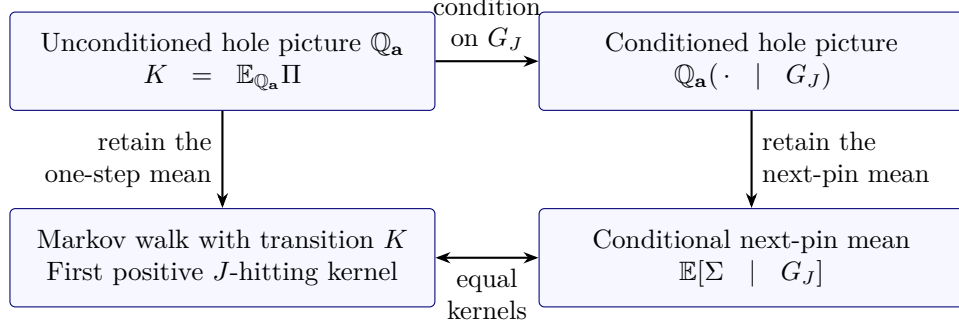


Figure 1: The first-moment reduction at a fixed exposure and a finite set J of pins on selected time cuts, with $\mathbb{Q}_{\mathbf{a}}(G_J) > 0$. The exterior-moment identity is used for the unconditioned law in the upper-left box; Proposition 4.3 proves that the two routes to the lower-right box agree. The lower-left walk resamples its transitions, whereas the upper-right law follows a single sampled permutation. The equality concerns the next-pin mean and does not identify these two path laws. When the exposure is averaged, its actual marginal still contains the factor $g_P(\mathbf{a})$ in (4.4).

4.3 Fresh turns and continuous boundary pins

Given an exposure $\mathbf{a}$ with $g_P(\mathbf{a}) > 0$, define the *fresh-turn walk* on holes as follows. Repeat periodically the hole sets, forced transports, and revealed endpoint restrictions. Between those prescribed changes, the walk jumps across every available slot edge at rate $1/2$. At a forced transport it follows that transport; at an endpoint completion it uses an independent uniform completion at the current site. Use fresh Poisson exchanges and completion choices in each turn. These prescriptions define a time-inhomogeneous Markov walk. On every bounded time interval it has finitely many jumps almost surely. For a starting point $i \in T$, write τ_T^+ for its first strictly positive visit to T , and τ_E for its first visit to E , allowing infinite hitting times.

Lemma 4.4 (Return identity with boundary pins). *For $\mathbb{P}_P$ -almost every exposure and every $i \in T$,*

$$\mathbb{P}_P(\text{next pin after } i \text{ lies in } T \mid \mathcal{H}) = \mathbb{P}_{\mathbf{a}}^{\text{walk}}(\tau_T^+ < \tau_E). \quad (4.11)$$

In particular the first moment in (4.1) is the $\mathbb{P}_P$ -average of the corresponding fresh-turn return probabilities in the exposed environment.

Proof. Fix an exposure for which Proposition 4.1 holds and $g_P(\mathbf{a}) > 0$. Choose increasing finite deterministic time sets $\mathcal{T}_n$ with dense union, and let $D_{0,n}$ contain all boundary slots at the times in $\mathcal{T}_n$. Put

$$E_n = E_f \cup D_{0,n}, \quad P_n = E_n \cup T.$$

The cuts used below include all times occurring in P_n , as well as time 0 if necessary. An extra unpinning cut merely subdivides an interval and does not change which cycles hit P_n .

Under $\mathbb{Q}_{\mathbf{a}}$, let G_n be the event that all hole cycles meet P_n , and set $g_n = \mathbb{Q}_{\mathbf{a}}(G_n)$. The events G_n increase. Almost surely the one-period hole picture has only finitely many random exchanges, in addition to the finitely many prescribed changes. Its spatial visits have positive residence time: independent Poisson jumps do not coincide with any prescribed change, and within-site completions do not change the site. The exposed histories avoid D_0 , so their cross jumps cannot create an instantaneous boundary excursion either. Every boundary visit by a hole

cycle is therefore detected by some cut in the dense union. There are finitely many cycles, whence

$$\mathbf{1}_{G_n} \uparrow \mathbf{1}_{G_P} \quad \mathbb{Q}_{\mathbf{a}}\text{-almost surely,} \quad g_n \uparrow g_P(\mathbf{a}) > 0. \quad (4.12)$$

In particular, $g_n > 0$ for all sufficiently large n . No invertibility is required for the earlier indices.

For such an n , Lemma 4.2 applies to the unconditioned layered hole permutation. With $J = P_n$, Proposition 4.3 identifies each next-pin probability under $\mathbb{Q}_{\mathbf{a}}(\cdot \mid G_n)$ with the corresponding first-hit probability for its matrix K . The latter is precisely the walk just defined, observed at the finitely many cuts: interval choices are independent, including on later passes through the same interval. Thus, if A_n is the event in a fixed hole picture that the next P_n -pin after i belongs to T ,

$$\frac{\mathbb{Q}_{\mathbf{a}}(G_n \cap A_n)}{g_n} = \mathbb{P}_{\mathbf{a}}^{\text{walk}}(\tau_T^+ < \tau_{E_n}). \quad (4.13)$$

On the picture side, a cycle through $i \in T$ returns to T after finitely many periods, regardless of which other pins it meets. Detection of boundary visits before that return shows that $\mathbf{1}_{A_n}$ decreases to the event A that the next full P -pin is in T . Together with (4.12) and bounded convergence, this gives

$$\mathbb{Q}_{\mathbf{a}}(G_n \cap A_n) \longrightarrow \mathbb{Q}_{\mathbf{a}}(G_P \cap A).$$

Dividing by the positive limiting denominator yields the return probability under $\mathbb{Q}_{\mathbf{a}}(\cdot \mid G_P)$.

For the walk side, use a single uncensored fresh-turn trajectory for all the tests. Its first positive T -hitting time τ_T^+ is then fixed, independently of n . The events on the right of (4.13) decrease. A path in their intersection has $\tau_T^+ < \infty$, since this is already part of each event. If that path had visited the boundary before τ_T^+ , the positive residence time of the visit would make one of the dense cuts detect it before τ_T^+ . Thus their intersection is exactly $\{\tau_T^+ < \tau_E\}$, up to a null set. Continuity from above gives convergence of the probabilities. Finally Proposition 4.1 identifies the limiting conditional picture law with the conditional law under $\mathbb{P}_P$. This proves (4.11). $\square$

4.4 Minimal hitting probabilities and return energy

We now ignore all pins of E except D_0 in the stopping rule for this walk. The exposed environment is kept fixed. This can only increase its probability of returning to T . Let $U = U(\mathbf{a})$ be its one-period column-forward transition matrix from the cut at θ to that same cut one period later, killed upon every visit to the boundary, including at its initial point. Boundary coordinates can be retained with value zero. With the post-jump convention at deterministic cuts, a period includes all changes after its starting cut and through its ending cut; in particular the within-site completion is included exactly once. All subsequent periods use fresh randomness.

We record the basic contraction property for later heat-flow estimates. For any subinterval, a realization of all exchanges, forced transports, and completions gives a bijection between its initial and final hole sets. Keeping only trajectories that avoid the boundary gives a partial permutation: its matrix has at most one entry equal to one in each row and in each column. The killed transition matrix C is the mean of these matrices. Hence

$$C_{ij} \geq 0, \quad \sum_i C_{ij} \leq 1, \quad \sum_j C_{ij} \leq 1, \quad \|Cv\|_2 \leq \|v\|_2. \quad (4.14)$$

Indeed Cauchy–Schwarz in each row gives $\sum_i |(Cv)_i|^2 \leq \sum_{i,j} C_{ij} |v_j|^2 \leq \sum_j |v_j|^2$. The same inequalities apply to adjoints and to additional killing. Row sums bound propagated values by one; column sums bound the total mass from a unit starting point. These are statements at a fixed exposure, without any assertion of symmetry of U or invariance of the exposure law under time reversal.

Lemma 4.5 (Minimal hitting function and return energy). *For almost every exposure under $\mathbb{P}_P$, let U be the boundary-killed one-period matrix just defined. There is a smallest nonnegative function f on its cut holes satisfying*

$$f = 1 \text{ on } T, \quad U^* f = f \text{ off } T. \quad (4.15)$$

It is given off T by the probability of ever reaching T on a future turn before boundary killing. In particular $0 \leq f \leq 1$, and f is zero on the boundary. With counting inner products,

$$\langle f, (I - U)f \rangle = \sum_{i \in T} (1 - (U^* f)_i), \quad m(1 - r) \geq \mathbb{E}_P \langle f, (I - U)f \rangle. \quad (4.16)$$

Both U and f in this expectation are functions of the same exposure, with the outer law in (4.4).

Proof. Set $f_0 = \mathbf{1}_T$ and recursively define

$$f_{n+1} = \mathbf{1}_T + \mathbf{1}_{H_\theta \setminus T} U^* f_n, \quad (4.17)$$

where the multiplication by the indicator is pointwise. Nonnegativity and the substochasticity of U^* show inductively that $0 \leq f_n \leq f_{n+1} \leq 1$. The Markov property identifies f_n , off T , with the probability of reaching T within the next n periods before killing. Thus its pointwise limit f has the asserted probability interpretation. There are finitely many coordinates, so passage to the limit in (4.17) gives (4.15). A boundary starting point is killed immediately, and consequently every f_n and f is zero there.

If h is any nonnegative solution of (4.15), then $h \geq f_0$ and induction in (4.17) gives $h \geq f_n$ for every n ; hence $h \geq f$. This proves minimality. In particular f is zero on any closed class that cannot reach T . No assumption of certain absorption by the boundary is needed.

Since T occurs at a single time modulo the period, a strictly positive return from that cut to T occurs after an integer number of periods. Conditioning on the first period therefore shows that the relaxed return probability from $i \in T$ is $(U^* f)_i$. By Lemma 4.4, the conditional probability $q_i(\mathbf{a})$ contributing to (4.1) is the return probability with all of E retained. Hence

$$q_i(\mathbf{a}) \leq (U^* f)_i \quad (i \in T).$$

The vectors are real, so transposition in the scalar quadratic form and harmonicity off T give

$$\begin{aligned} \langle f, (I - U)f \rangle &= \langle (I - U^*)f, f \rangle \\ &= \sum_{i \in T} (1 - (U^* f)_i) \leq \sum_{i \in T} (1 - q_i(\mathbf{a})). \end{aligned}$$

Averaging under the actual exposure marginal gives (4.16). Measurability causes no additional limiting issue: finite-time transition probabilities are measurable functions of the finite-jump exposure, and f is the increasing limit of the finite-horizon functions in (4.17). $\square$

For the estimates that follow, write $\eta = \mathbf{1}_T$ and $g = f - \eta$. Then $0 \leq g \leq 1$, $\langle \eta, g \rangle = 0$, and both functions have zero boundary values. Lemma 4.5 shows that the desired scale of quadratic loss is m : for any $\varepsilon \geq 0$,

$$\mathbb{E}_P \langle f, (I - U)f \rangle \geq m(1 - \varepsilon) \implies r \leq \varepsilon.$$

The simultaneous pin test then bounds the old-pin avoidance probability by $(2\varepsilon)^m$. The next sections obtain this lower bound with a small error from the spatial distribution of holes and the corresponding heat flow.

5 Sparse obstacles and local diffusion

The return-energy problem requires two spatial estimates in each exposed environment. Mass initially on T must spread through nearby holes, and its pairing, after propagation, with the remainder $g = f - \mathbf{1}_T$ of the hitting function must be bounded by the energy of that propagated remainder. Short connections near each starting point supply the first estimate; flows through the rest of the lattice will supply the second. We now derive the geometric conditions for both from the sparse pin-test induction. This section bounds the probability that these conditions fail and proves the local Nash inequality. Section 6 constructs the flows, and Section 7 uses both estimates to close the induction.

5.1 The induction input and blocking between cuts

Fix throughout

$$\alpha = \frac{1}{100}, \quad \nu = \frac{1}{100d}, \quad a = \frac{11}{10}. \quad (5.1)$$

All spatial distances in this and the following two sections are in the supremum norm, and $B_h(x) = \{z \in \mathbb{Z}^d : |z - x|_\infty \leq h\}$. A finite collection of slot points with spatial coordinates $(x_i)_{i \in A}$ is called *sparse* if

$$\#\{i \in A : x_i \in B_h(z)\} \leq L(1+h)^\alpha \quad (z \in \mathbb{Z}^d, h \geq 0). \quad (5.2)$$

Repeated spatial coordinates are counted with their slot multiplicities. The constant $L \geq \max\{2d, \ell, 4\}$ is fixed independently of all scales. For a set of distinct sites, sparseness means the same inequality with one point per site. Every subset of a sparse collection is sparse.

Let $0 < p < 1$, put

$$s = p^{-a}, \quad R = \lceil C_* s \rceil, \quad \beta \geq 5s, \quad (5.3)$$

and choose a finite circular time grid $\mathcal{G} \subset \mathbb{R}/\beta\mathbb{Z}$ containing 0, whose successive gaps are at most p^{4d} . Here $C_* \geq 1$ is a fixed constant, to be chosen for the localization estimate. Write $\mathcal{V}_\mathcal{G} = V_\Lambda \times \mathcal{G}$. For $D = D_0 \cup F$, where $F \subseteq \mathcal{V}_\mathcal{G}$, let $\mathsf{l}_p(D)$ denote the assertion

$$\mathbb{P}_D(A \subseteq \mathcal{B}_D) \leq p^{|A|} \quad \text{for every nonempty sparse set } A \subseteq \mathcal{V}_\mathcal{G} \setminus D \text{ at a single grid time.} \quad (5.4)$$

There are only finitely many grid points outside D_0 . Thus one may prove these assertions by downward induction on the number of pins, starting from the vacuous assertion when every grid point is pinned. In a step with pin set E and test T , the set $P = E \cup T$ has strictly more pins. In the present section we assume $\mathsf{l}_p(P)$; we do not assume $\mathsf{l}_p(E)$. Under this hypothesis we will seek a bound $r(E, T) \leq \varepsilon(p)$ with $\varepsilon(p) = o(p)$. Proposition 3.1 then yields the required $p^{|T|}$ bound for the test under $\mathbb{P}_E$, once $2\varepsilon(p) \leq p$.

Expose $\mathcal{B}_P$ under $\mathbb{P}_P$ as in Proposition 4.1. A site is *blocked* at time t if none of its ℓ slots is a hole. Denote the blocked set by $\mathcal{O}_t$. Extend the environment to $\mathbb{Z}^d$ by making all slots outside Λ available. Boundary sites also have all slots available because $D_0 \subseteq P$. In particular $\mathcal{O}_t$ is finite and is disjoint from the inner boundary.

Sparse growth serves two purposes. A blocked path crossing a large spatial scale will supply a separated collection satisfying (5.2), so the restricted induction hypothesis can still detect large obstacles. Later, the same growth bound on the test points will make overlaps between their decaying flows summable. The component estimate below and the energy estimate in Proposition 6.4 implement these two uses.

At a grid time q , the induction input gives

$$\mathbb{P}_P(A \subseteq \mathcal{O}_q) \leq p^{|A|} \quad \text{for every prescribed sparse set of distinct sites } A. \quad (5.5)$$

Indeed, choose one fixed slot at each site. If any chosen slot is pinned, or any site is outside the volume, blocking is impossible. Otherwise the chosen slot points form an admissible test in $\mathfrak{l}_p(P)$.

Lemma 5.1 (Blocking at a deterministic time). *Assume $\mathfrak{l}_p(P)$, and let t be any deterministic time. There is a constant $C_0 = C_0(d, \ell)$, uniform in $p \leq 1$, the volume, the period, the grid, and P , such that, for every prescribed finite set $A \subset \mathbb{Z}^d$ of distinct sites,*

$$\mathbb{P}_P(A \subseteq \mathcal{O}_t) \leq C_0 p^{2d} \quad \text{if } |A| = 2d, \quad (5.6)$$

$$\mathbb{P}_P(A \subseteq \mathcal{O}_t) \leq (C_0 \sqrt{p})^{|A|} \quad \text{if } A \text{ is a nonempty sparse set of distinct sites.} \quad (5.7)$$

Proof. Let q be the preceding grid time and $I = (q, t]$, interpreted cyclically, so that $|I| \leq p^{4d}$. On-site permutations do not change the number of holes at a site. Consequently a site blocked at t was either blocked at q or has an incident inter-site mark in I . This remains true when I crosses the time seam. By Lemma 2.4, the expected number of marks in the edge star of a prescribed site during I is at most $2d\ell^2|I|$. If all $2d$ prescribed sites are blocked at t , either they were all blocked at q , or one of these finitely many edge stars contains a mark. Their set is sparse since $L \geq 2d$. The union bound and (5.5) give $p^{2d} + Cp^{4d} \leq Cp^{2d}$.

Now let $n = |A|$ and assume A is sparse. If every site of A is blocked at t , either at least $\lceil n/2 \rceil$ of its sites were blocked at q , or at least $\lceil n/2 \rceil$ have an incident mark in I . For the first alternative, summing (5.5) over subsets costs at most $2^n p^{\lceil n/2 \rceil}$. For the second, one of the two lattice parities supplies at least $k = \lceil n/4 \rceil$ sites with incident marks. Edge stars of distinct sites of the same parity are disjoint. If N_z counts marks in the star of z during I , then for any such prescribed k -set A' , the product $\prod_{z \in A'} N_z$ counts choices of k distinct marks. The factorial bound therefore yields

$$\mathbb{P}_P(N_z \geq 1 \text{ for all } z \in A') \leq \mathbb{E}_P \prod_{z \in A'} N_z \leq (2d\ell^2|I|)^k.$$

Summing over at most 2^{n+1} possible choices bounds the second alternative by $2^{n+1}(Cp^{4d})^{\lceil n/4 \rceil}$. As $4d\lceil n/4 \rceil \geq dn \geq n/2$, both alternatives are bounded together by $(C_0\sqrt{p})^n$, after enlarging C_0 . These are separate event bounds combined by a union bound; no independence between old blocking and intervening marks is required. $\square$

5.2 Admissible binary witnesses and component tails

Join two blocked sites when their supremum distance is at most 8. All components and paths of blocked sites below use this adjacency. For $z \in \mathcal{O}_t$, let $\mathcal{C}_t(z)$ be its component and set

$$\text{rad}_z \mathcal{C}_t(z) = \max_{w \in \mathcal{C}_t(z)} |w - z|_\infty.$$

The next lemma gives a tail uniform even as the bootstrap parameter is subsequently decreased.

Lemma 5.2 (Uniform component tail). *There are $p_0 > 0$ and $c, C, \gamma > 0$, depending only on d, ℓ and the fixed sparse-growth choices, such that whenever $0 < p \leq p_0$ and $\mathfrak{l}_p(P)$ holds,*

$$\mathbb{P}_P(z \in \mathcal{O}_t, \text{rad}_z \mathcal{C}_t(z) \geq h) \leq C \exp(-ch^\gamma) \quad (z \in \mathbb{Z}^d, h \geq 1) \quad (5.8)$$

at every deterministic time t . The constants c, C, γ are fixed once p_0 is fixed and work simultaneously for all $p \leq p_0$.

Proof. Fix an integer $b > 100$ so large that $b^\alpha > 4$, and put $h_j = 16b^j$. We first describe exactly which site lists will be used as witnesses. Their separated binary-tree structure is also used in multiscale cascading-event arguments; see, for example, [34, Section 3]. Here the leaf probabilities will come directly from (5.7). A depth- j witness rooted at z is an assignment of sites x_w to the vertices w of a rooted ordered binary tree of depth j , with $x_\emptyset = z$. If a vertex w has $k \geq 1$ levels below it, require

$$x_{w0} = x_w, \quad x_{w1} \in B_{h_k+8}(x_w), \quad \text{dist}_\infty(\mathcal{L}_{w0}, \mathcal{L}_{w1}) \geq h_k/4, \quad (5.9)$$

where $\mathcal{L}_{wa}$ is the set of leaf sites descended from child wa . Call such an assignment *admissible*. Its leaves are distinct by the separation condition. Depth zero has just the prescribed root as its leaf.

A blocked crossing from z to distance at least h_j produces an admissible witness all of whose leaves are blocked. For $j = 0$ use the root alone. For $j \geq 1$, truncate the crossing at its first exit to distance h_j ; it is contained in $B_{h_j+8}(z)$. One child crossing starts at z . The other starts at the first site whose distance from z is at least $h_j/2$, hence at distance at most $h_j/2 + 8$. The remaining part of the parent crossing reaches distance at least $h_j/2 - 8$ from this second start. Each child can thus be truncated at its first exit to distance h_{j-1} , and is contained in its start-centered ball of radius $h_{j-1} + 8$. The distance between these two balls is at least

$$h_j/2 - 2(h_{j-1} + 8) \geq h_j/4.$$

Recurse on the two truncated child crossings. Every descendant leaf remains in its child's truncated crossing, so all the separation conditions in (5.9) hold.

Every admissible witness has a sparse leaf set. Indeed, for $1 \leq k \leq j$, a ball of diameter strictly less than $h_k/4$ meets at most one child at every split of scale at least h_k . It therefore contains at most 2^{k-1} leaves. If $h_{k-1}/8 \leq h < h_k/8$, this bound is at most $L(1+h)^\alpha$, since

$$L(1+h)^\alpha \geq L(2b^{k-1})^\alpha \geq 2^{k-1}.$$

For $h < h_0/8$ there is at most one leaf, and for $h \geq h_j/8$ the same calculation allows all 2^j leaves. The depth-zero case is immediate. This verifies (5.2) for all centers and real radii, including the scale endpoints.

Let A_j be an upper bound on the number of admissible assignments rooted at any prescribed site. At the top split the first child root is fixed and the second has at most $(C_d h_j)^d$ choices. Ignoring restrictions between the two child assignments gives the numerical overcount

$$A_0 = 1, \quad A_j \leq (C_d h_j)^d A_{j-1}^2.$$

In particular

$$2^{-j} \log A_j \leq d \sum_{k=1}^j 2^{-k} \log(C_d h_k) \leq \log C_b < \infty, \quad A_j \leq C_b^{2^j}.$$

Only admissible assignments enter the probability union bound. The unrestricted product used to bound their number need not itself consist of sparse witnesses.

For each admissible assignment, Lemma 5.1 bounds the probability that all its leaves are blocked by $(C_0 \sqrt{p})^{2^j}$. Choose $p_0 \leq 1$ once and for all so that $C_b C_0 \sqrt{p_0} \leq e^{-2}$. For every $p \leq p_0$,

$$\mathbb{P}_P(z \in \mathcal{O}_t, \text{rad}_z \mathcal{C}_t(z) \geq h_j) \leq \exp(-2 \cdot 2^j). \quad (5.10)$$

For $h \geq 16$, choose $j \geq 0$ with $h_j \leq h < h_{j+1}$. Then $2^j \geq (h/(16b))^\gamma$, where $\gamma = \log_b 2 > 0$. This proves (5.8); increasing C covers $1 \leq h < 16$. No constant in this last step changes when p is decreased further. $\square$

5.3 Regular anchors

We now separate the geometry near the starting point of a localized mass from the geometry farther away. Nearby, fewer than $2d$ blocked sites in each fixed-radius ball will preserve short lattice connections; these connections transfer the lattice Nash inequality to the holes. Farther away, a sublinear bound on component radii will control the displacement of the routes used for flows to infinity in Section 6. The first condition gives local diffusion, and the second permits the global energy comparison. Let M_* be a fixed positive integer. Its required size will be specified below, independently of p .

Definition 5.3 (Regular anchor). For a spatial anchor $x \in \mathbb{Z}^d$, call time t *regular for x* at scale R if both conditions hold:

1. For every $z \in B_{11R}(x)$, $\#(\mathcal{O}_t \cap B_{M_*}(z)) < 2d$.
2. For every $z \in \mathcal{O}_t \setminus B_{10R}(x)$, $\text{rad}_z \mathcal{C}_t(z) \leq (1 + |z - x|_\infty)^\nu$.

For $i \in T$, “regular for i ” means regular for its spatial coordinate x_i .

Proposition 5.4 (Probability of irregularity). *Fix M_* . Under $\mathbb{I}_p(P)$, for all $0 < p \leq p_0$, all deterministic times t , and all spatial anchors x ,*

$$\mathbb{P}_P(t \text{ is not regular for } x) \leq \delta := CR^d p^{2d} + C \exp(-cR^\xi), \quad \xi = \nu\gamma > 0. \quad (5.11)$$

Here $R \geq 1$, and the constants may depend on M_* but are uniform in $p \leq p_0$, volume, period, grid, pin set and anchor.

Proof. There are at most CR^d centers in $B_{11R}(x)$, and a ball of radius M_* has a fixed number of $2d$ -element subsets. The union bound and (5.6) therefore bound failure of the first condition by $CR^d p^{2d}$. By Lemma 5.2, failure of the second has probability at most

$$C \sum_{z: |z-x|_\infty > 10R} \exp(-c(1 + |z - x|_\infty)^\nu) \leq C \exp(-c'R^{\nu\gamma}).$$

For the last inequality, count at most $C(1 + n)^{d-1}$ sites at distance n and absorb this polynomial and the tail summation into half of the exponential. Since $\nu\gamma > 0$ is fixed, the resulting constants are uniform. Add the two bounds. $\square$

We will use (5.11) only at prescribed times and after integration in time. It asserts neither independence between slices nor regularity simultaneously at every time.

5.4 Local connections and the Nash inequality

The remaining statements in this section are deterministic. At a fixed time let $\mathcal{H}_t$ be the graph whose vertices are the available slots on the extended lattice and whose edges join every pair of available slots above nearest-neighbor sites. For a real function v on its vertices define

$$\mathcal{D}_t(v) = \frac{1}{2} \sum_{\{i,j\} \in E(\mathcal{H}_t)} (v_i - v_j)^2. \quad (5.12)$$

Edges in this sum are unoriented and counted once. Norms use counting measure on the available slots. When a function describes killed mass, its values at the physical boundary and outside the volume are zero. Thus (5.12) includes the edges into the zero-valued boundary, as required by Dirichlet killing.

Lemma 5.5 (Uniform local connectivity). *Let $d \geq 2$ and $K \geq 1$ be fixed. There is an integer $\rho = \rho(d, K) \geq K + 1$ such that the following holds. If $u, v \in \mathbb{Z}^d$, $|u - v|_\infty \leq K$, and $F \subset \mathbb{Z}^d$ satisfies*

$$u, v \notin F, \quad \#(F \cap B_\rho(u)) < 2d,$$

then a nearest-neighbor path from u to v avoids F , lies in $B_\rho(u)$, and has length at most $(2\rho + 1)^d$.

Proof. First suppose F is a set of fewer than $2d$ sites in the whole lattice. The complement of a box containing F is connected when $d \geq 2$, so $\mathbb{Z}^d \setminus F$ has only one possible infinite component. It has no finite component: from a vertex of such a component the first exit along each of the $2d$ positive and negative coordinate rays must lie in F , giving $2d$ distinct deleted sites. Hence $\mathbb{Z}^d \setminus F$ is connected.

The confinement radius can be uniform when the two endpoints have bounded separation. Otherwise, after translating the first endpoint to 0, there would be sets F_n with $|F_n| < 2d$ and surviving endpoints $v_n \in B_K(0)$, admitting no path inside $B_n(0)$. Pass to a subsequence with $v_n = v$ and with the indicators of F_n converging pointwise on $\mathbb{Z}^d$. The limiting set F_∞ has fewer than $2d$ sites and avoids both endpoints. By the preceding paragraph there is a finite path from 0 to v avoiding F_∞ . For all sufficiently large n , this path lies in $B_n(0)$ and avoids F_n , a contradiction. Choose a uniform radius $\rho \geq K + 1$.

For the asserted local version apply this conclusion to $F' = F \cap B_\rho(u)$, a set of fewer than $2d$ sites in the whole lattice. The resulting path lies in $B_\rho(u)$, where F' and F agree. Erasing loops gives the claimed length bound. $\square$

We may now make the local radius choice without reference to p . Fix

$$K_{\text{conn}} = 8(2d - 2) + 12, \quad M_* \geq \max\{8(2d), \rho(d, K_{\text{conn}}) + 2\}. \quad (5.13)$$

We henceforth require R to exceed these fixed local radii, which is achieved by decreasing the final p . At a regular anchor, Lemma 5.5 is now available for surviving sites within separation K_{conn} , whenever its first endpoint lies in $B_{11R}(x)$. In particular, it applies to every short connection used below. This choice also serves the local source connections in Section 6.

The next estimate uses Nash's norm-energy inequality and its diffusion mechanism [31, Part I]. The transfer to the available slots uses path comparison of Dirichlet forms, a method developed systematically by Diaconis and Saloff-Coste [9, Theorem 2.1]. We supply the comparison for the present obstacles and zero extension explicitly.

Proposition 5.6 (Local Nash inequality). *Fix $d \geq 3$, ℓ , and M_* as in (5.13). Let $\mathcal{H}_t$ be an available-slot graph with at least one and at most ℓ slots at every unblocked site, and suppose t is regular for x at scale $R \geq M_* + K_{\text{conn}} + 3$. Every nonnegative function v supported on slots above $B_R(x)$ satisfies*

$$\|v\|_2^{2+4/d} \leq C \mathcal{D}_t(v) \|v\|_1^{4/d}, \quad (5.14)$$

where C depends only on d, ℓ and the fixed local radii.

Proof. We transfer v to a lattice function while controlling energy, then prove the lattice inequality by averaging. At every surviving site z , choose an available slot maximizing v , and let $V(z)$ be this maximum. Every blocked site in $B_{R+3}(x)$ has a surviving nearest neighbor: otherwise its $2d$ neighbors alone would violate the first regularity condition in the ball centered there. Choose one

such neighbor $\phi(z)$ and define

$$w(z) = \begin{cases} V(z), & z \notin \mathcal{O}_t, \\ V(\phi(z)), & z \in \mathcal{O}_t \cap B_{R+3}(x), \\ 0, & z \in \mathcal{O}_t \setminus B_{R+3}(x). \end{cases}$$

The function w is supported in $B_{R+1}(x)$. Since a site's value is copied only by its nearest neighbors,

$$\|v\|_2^2 \leq \ell \|w\|_2^2, \quad \|w\|_1 \leq (2d+1)\|v\|_1. \quad (5.15)$$

For a lattice edge $\{z, z'\}$ with a nonzero endpoint value, both endpoints belong to $B_{R+2}(x)$. Each of $w(z), w(z')$ is the value at a chosen maximizing slot over a surviving site, say u, u' , at distance at most one from the corresponding endpoint. Thus $|u - u'|_\infty \leq 3$ and $u, u' \in B_{R+3}(x)$. Lemma 5.5 joins them through surviving sites within a fixed radius of u , with a fixed length bound. Lift this path using the chosen maximizing slot at every site. Every lifted edge is available because all pairs of available slots across a lattice edge are joined. If $u = u'$, the difference is zero. Otherwise telescoping and Cauchy–Schwarz show that $(w(z) - w(z'))^2$ is at most a fixed constant times the sum of $(v_i - v_j)^2$ along the lifted path. For any fixed slot edge, only a bounded number of the original lattice edges can have a path using it: all of their endpoints lie within a fixed radius of that edge. Hence, summing,

$$\sum_{\{z, z'\}: |z-z'|_1=1} (w(z) - w(z'))^2 \leq C\mathcal{D}_t(v). \quad (5.16)$$

For completeness, write $\mathcal{E}_{\mathbb{Z}^d}(w)$ for the sum on the left of (5.16). For an integer $k \geq 1$, let $A_k w$ be the average of translates of w by the points of $\{0, \dots, k-1\}^d$. Telescoping a translate along coordinate directions, followed by Cauchy–Schwarz, gives

$$\|w - A_k w\|_2^2 \leq C_d k^2 \mathcal{E}_{\mathbb{Z}^d}(w).$$

The averaging kernel has counting 2-norm $k^{-d/2}$; the convolution inequality therefore gives $\|A_k w\|_2^2 \leq k^{-d} \|w\|_1^2$. Consequently

$$\|w\|_2^2 \leq C_d k^2 \mathcal{E}_{\mathbb{Z}^d}(w) + 2k^{-d} \|w\|_1^2.$$

If $w \neq 0$, choose $k = \lceil (4\|w\|_1^2/\|w\|_2^2)^{1/d} \rceil$. The second term is at most $\|w\|_2^2/2$, and $\|w\|_1^2/\|w\|_2^2 \geq 1$ bounds the effect of rounding. Absorption yields

$$\|w\|_2^{2+4/d} \leq C_d \mathcal{E}_{\mathbb{Z}^d}(w) \|w\|_1^{4/d}.$$

For $w = 0$ the inequality is immediate. Finally use (5.15) and (5.16) to obtain (5.14). $\square$

The order of choices is now explicit. First fix d, ℓ , the exponents in (5.1), the witness scale ratio b , and L . Next fix the preliminary p_0 and the uniform tail constants in Lemma 5.2. Fix the connectivity radii and M_* by (5.13); this fixes the constants in Proposition 5.4 and Proposition 5.6. Choose C_* for localization, and only then decrease $p \leq p_0$ so that R exceeds the fixed radii and the leakage error is small enough to close the induction. Thus no tail constant deteriorates as the final p is decreased. The next section uses the same regularity conditions to construct flows to infinity, complementing the local diffusion bound proved here.

6 Flows around the blocked components

Fix one time slice t . The graph $\mathcal{H}_t$ consists of the available slots, including all ℓ slots at every site outside Λ , with all available slot pairs across each nearest-neighbor site edge. Write $\mathcal{Z} = \mathcal{O}_t$ for its finite blocked set. We will represent masses near regular anchors as divergences of flows with controlled energy. This will bound their pairing with a function by that function's Dirichlet energy, the estimate needed for the cross term in Section 7. The divergence and energy viewpoint is the classical electrical-network flow method; see [11, Sections 1.3.5 and 2.4.6]. The geometric work here is to retain a quantitative decay bound after routing through this dependent obstacle configuration.

Give every unoriented slot edge e a reference orientation (e^-, e^+) . A flow is a real function J on these reference edges; its divergence is outgoing minus incoming flow. Thus a signed unit path from a to b has divergence $\mathbf{1}_{\{a\}} - \mathbf{1}_{\{b\}}$. Write $\text{dist}(e, y) = \min\{|x - y|_\infty : x \text{ is an endpoint site of } e\}$. Constants in this Section depend only on d, ℓ and the fixed geometric constants in Section 5, never on the slice or volume.

The construction begins with a full-lattice unit flow whose strength at distance h from its source is $O((1 + h)^{1-d})$. We replace each oriented lattice edge by a path through available slots, with endpoints given by one application of a map on the original vertices. A short additional path then restores the source at the specified slot. The geometry below bounds the displacement of the replacement paths, and hence the number of original edges that can contribute to a given available edge. The resulting decay will let us add flows from a sparse collection of sources with controlled total energy.

6.1 Filled hulls and their surviving boundaries

We first thicken each blocked component to obtain clearance for routes around it. After filling the finite pockets of its complement, we will show that the exterior vertex boundary is connected by sup-norm steps of length at most one. These steps can be replaced by unblocked nearest-neighbor paths. The filled sets may be nested; their inclusion-maximal members will give disjoint regions on which to define the vertex map.

Let C range over the components of $\mathcal{Z}$ for steps of sup norm at most eight. Set

$$A_C = \{z \in \mathbb{Z}^d : \text{dist}(z, C) \leq 4\}, \quad H_C = \mathbb{Z}^d \setminus U_C,$$

where U_C is the infinite nearest-neighbor component of $\mathbb{Z}^d \setminus A_C$. Such an infinite component is unique: the exterior of a box containing A_C is connected when $d \geq 2$. Thus H_C is A_C with its finite complementary components filled in. The exterior vertex boundary of H_C is

$$S_C = \{v \notin H_C : \text{some } u \in H_C \text{ satisfies } |u - v|_1 = 1\}.$$

The exterior-boundary connectivity in the next lemma is the lattice case of Timár's boundary-connectivity result [35, Lemmas 1–2]. We include the plaquette proof together with the separate hull and clearance estimates needed for the routes.

Lemma 6.1 (Hull topology and clearance). *Let $d \geq 2$ and let $\mathcal{Z} \subset \mathbb{Z}^d$ be finite. The sets A_C are pairwise disjoint and nearest-neighbor connected. Each H_C is finite, is nearest-neighbor connected, has connected complement, and is contained in the coordinate bounding box of A_C . Any two hulls are disjoint or one contains the other.*

For every edge $\{u, v\}$ with $u \in H_C$, $v \notin H_C$,

$$\text{dist}(u, C) = 4, \quad \text{dist}(v, C) = 5. \quad (6.1)$$

Both endpoints are unblocked. The set S_C is connected by steps of sup norm at most one. Each such step can be replaced by a nearest-neighbor path of unblocked sites, contained in the unit enlargement of the bounding box of A_C . Consequently any two points of S_C can be joined by a finite path of unblocked sites in that enlargement.

Proof. Distinct components have mutual distance at least nine, so their radius-four thickenings are disjoint. The radius-four lattice cubes about successive points of a component intersect; their union is connected. Every finite complementary component has a neighbor in A_C , which proves that filling preserves connectivity. A vertex outside the coordinate bounding box has a coordinate ray avoiding that box, so it belongs to U_C . This proves finiteness and the claimed containment.

For laminarity, take two disjoint connected thickenings $A_C, A_{C'}$. The second lies in a single component of the complement of the first. If it lies in a finite component, then U_C is an infinite connected subset of the complement of $A_{C'}$, whence $U_C \subset U_{C'}$ and $H_{C'} \subset H_C$. The reversed containment is treated identically. In the remaining case each thickening lies in the infinite complementary component of the other. Then H_C avoids $A_{C'}$ and is connected; since it contains $A_C \subset U_{C'}$, it lies wholly in $U_{C'}$. Hence H_C and $H_{C'}$ are disjoint.

We next prove connectivity of the exterior boundary without a planar separation argument. Let $\mathcal{E}_C$ be the edges between H_C and U_C , and join two of these edges when they lie on an elementary lattice plaquette. Every plaquette meets $\mathcal{E}_C$ an even number of times, and all cut edges of one plaquette belong to the same component of this edge relation. Thus each component separately meets every plaquette evenly. It then meets every finite closed lattice walk evenly: modulo two, such a walk is a sum of elementary plaquettes, as follows by commuting successive steps in different coordinate directions and canceling inverse steps. If the cut had two components, take one edge from each, join their inner endpoints within H_C and their outer endpoints within U_C . The resulting closed walk uses exactly one cut edge from each chosen component, a contradiction. The cut is therefore connected by plaquettes. The outer endpoints of cut edges on a common plaquette have sup distance at most one. This proves the asserted connectivity of S_C .

An inner cut endpoint u must belong to A_C : otherwise a complementary pocket containing u would be adjacent to the infinite component through v , which is impossible. Since $v \notin A_C$ and $|u - v|_\infty = 1$, the distances to C are exactly four and five. For another component C' , separation gives $\text{dist}(v, C') \geq 4$ and $\text{dist}(u, C') \geq 5$. To interpolate a sup step between two points of S_C , change the differing coordinates one at a time. Every intermediate vertex is within sup distance one of its starting point, hence has distance at least four from C and at least three from every other component. All these vertices survive. They also remain in the coordinate bounding box of S_C , which is contained in the unit enlargement stated above. $\square$

Keep only the inclusion-maximal hulls and denote their family by $\mathfrak{H}$. They are pairwise disjoint and contain all blocked sites. Choose one representative $r_H \in S_C$ for each $H = H_C \in \mathfrak{H}$, and define a map on sites by

$$\rho(z) = \begin{cases} r_H, & z \in H \in \mathfrak{H}, \\ z, & z \notin \bigcup_{H \in \mathfrak{H}} H. \end{cases} \quad (6.2)$$

All images are unblocked. The map is applied once. A representative can lie in another hull, so no identity $\rho(r_H) = r_H$ is assumed.

For each oriented lattice edge (u, v) , join $\rho(u)$ to $\rho(v)$ as follows. If both endpoints are in the same hull, use the empty path. If neither is in a hull, use the edge itself. If only u belongs to a hull H , then $v \in S_C$ and a path along its surviving boundary joins r_H to v ; reverse this prescription when only v is in a hull. Finally, if $u \in H$ and $v \in H'$ for distinct maximal hulls, both u and v survive and the signed route is

$$r_H \longrightarrow v \longrightarrow u \longrightarrow r_{H'}. \quad (6.3)$$

The first portion uses the exterior boundary of H , the last the exterior boundary of H' , and the middle portion traverses the original edge in reverse. Indeed $v \in S_C$ and $u \in S_{C'}$. Erase loops from the resulting walk, leaving a simple path with the same endpoints. This path stays in the enlarged bounding boxes of the hulls incident to the original edge, together with that edge. Figure 2 shows an exterior-boundary detour and the adjacent-hull case in which the one-use convention and the sign of the middle portion matter.

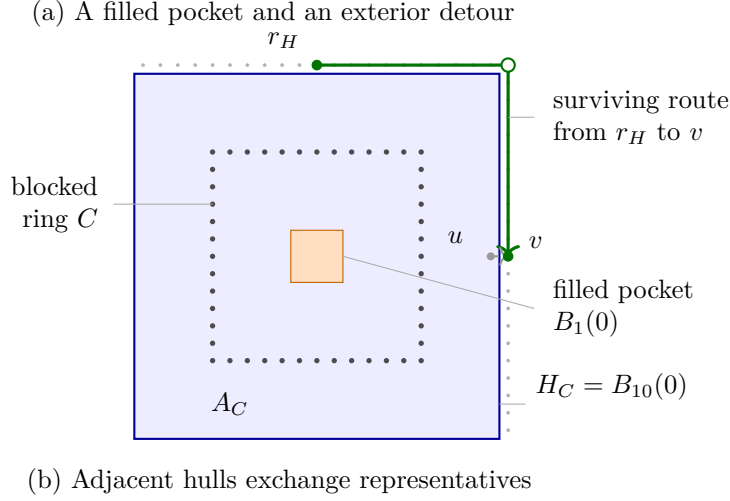


Figure 2: Replacing lattice edges by surviving paths. In (a), a two-dimensional example, $C = \{z : |z|_\infty = 6\}$ is a blocked ring. Its radius-four thickening is $A_C = \{z : 2 \leq |z|_\infty \leq 10\}$; filling the surviving pocket $B_1(0)$ gives $H_C = B_{10}(0)$. The original edge $u \rightarrow v$, with $u = (10, 0)$ and $v = (11, 0)$, is replaced by the indicated path from $r_H = (0, 11)$ to v . The open corner is a surviving interpolation vertex outside S_C : a nearest-neighbor route need not remain in S_C itself. In (b), the only blocked sites are 0 and $9e_1$. Their filled radius-four hulls are the two displayed lattice cubes, shown schematically in a coordinate slice; the transverse scale is compressed. Both $u = 4e_1$ and $v = 5e_1$ survive. Choosing $r_{H_0} = v$ and $r_{H_1} = u$ sends the oriented original edge $u \rightarrow v$ to the reverse edge $v \rightarrow u$. This is the case of (6.3) in which the two boundary portions are empty. Applying ρ again would undo this required image exchange. Panel (a) illustrates the definitions in $d = 2$; the hull and boundary properties in every $d \geq 2$ are proved in Lemma 6.1.

6.2 A decaying unit flow from every nearby slot

The topology gives routes for all lattice edges. Regularity now bounds their displacement, which will bound the number of routes using any one target edge.

Proposition 6.2 (Decay of a routed unit flow). *Let $d \geq 3$, $\nu = 1/(100d)$, and suppose the anchor x is regular in the sense of Definition 5.3. Fix the local constant M_* as in (5.13) and take R sufficiently large in terms of those fixed constants. For every available slot a at any site $y \in B_R(x)$ there is a flow J_a on the available-slot graph such that*

$$\operatorname{div} J_a = \mathbf{1}_{\{a\}}, \quad |J_a(e)| \leq C(1 + \operatorname{dist}(e, y))^{-k}, \quad k = d - 1 - d\nu. \quad (6.4)$$

In particular, $\sum_e J_a(e)^2 < \infty$.

Proof. First we quantify the hull sizes. A blocked component meeting $B_{10R}(x)$ has fewer than $2d$ vertices. Otherwise, starting at a vertex there, a connected exploration would find $2d$ vertices within $2d - 1$ component steps, all in a ball of radius $8(2d - 1) \leq M_*$, contradicting regularity. Such a component has diameter at most $8(2d - 2)$, and its hull has diameter at most

$$D_{\text{loc}} = 8(2d - 2) + 8.$$

For a component avoiding $B_{10R}(x)$, let $b_C = \min_{z \in C} |z - x|_\infty > 10R$. Applying remote regularity at a minimizer gives

$$\operatorname{diam} H_C \leq 2(1 + b_C)^\nu + 8. \quad (6.5)$$

This also bounds the diameter of its coordinate bounding box. Since $\nu < 1$, increasing a fixed lower threshold for R makes the right side, plus any of the fixed enlargements used above, at most $b_C/10$. All sites of these enlarged boxes then have distance comparable to b_C from every $y \in B_R(x)$. In particular, no such remote box contains y .

Apply the routes following (6.2), using fixed choices of representatives and paths. The preceding size bounds show that the route of a lattice edge e lies within distance

$$C(1 + \operatorname{dist}(e, y))^\nu \quad (6.6)$$

of that edge. For a local hull this is the fixed bound $D_{\text{loc}} + 2$. For each remote hull incident to e , an endpoint of e belongs to its bounding box, so its b_C is comparable to $1 + \operatorname{dist}(e, y)$; now use (6.5). The same reasoning applies when the edge joins two hulls. These comparisons are uniform over all injection sites $y \in B_R(x)$.

Choose one canonical available slot $c(z)$ at each unblocked site z . Lift every routed site path to these canonical slots. All its edges exist because every pair of available slots across a lattice edge is present. The map of the original lattice vertices into slots is $\psi(z) = c(\rho(z))$. Its preimages are finite: outside the finitely many finite hulls it is injective, and only those hulls can supply additional preimages.

We also need a bounded path from the specified slot a to $\psi(y)$. If y is in a hull, that hull is local and the separation between y and $\rho(y)$ is at most $D_{\text{loc}} + 1$. Both sites are unblocked, so Lemma 5.5 joins them through survivors in a fixed neighborhood of y . Its assumptions hold because $y \in B_R(x)$, the neighborhood has fixed radius, and R dominates that radius. Lift this path, starting at a . If $\rho(y) = y$ but $a \neq c(y)$, use a two-edge path through a surviving neighbor of y . Such a neighbor exists, since blocking all $2d$ neighbors would violate local regularity. Thus in all cases there is a simple slot path Q_a from a to $\psi(y)$ of bounded length and displacement.

The choice $K_{\text{conn}} \geq 8(2d - 2) + 12$ in the local-connectivity construction covers these requirements before M_* , R , and finally p are fixed.

Start with an explicit unit flow F_y on the full lattice. In the positive orthant based at y , give each site $y + n$, $n \in \mathbb{Z}_{\geq 0}^d$, $|n|_1 = N$, mass

$$a_N = \binom{N + d - 1}{d - 1}^{-1},$$

and send the fraction $(n_j + 1)/(N + d)$ along its edge in direction e_j . Put zero on all other edges. At a site $y + n$ of level $N \geq 1$, the incoming mass is

$$a_{N-1} \frac{\sum_j n_j}{N + d - 1} = a_{N-1} \frac{N}{N + d - 1} = a_N,$$

which equals its outgoing mass. The outgoing mass at y is one. Consequently

$$\text{div } F_y = \mathbf{1}_{\{y\}}, \quad |F_y(e)| \leq C(1 + \text{dist}(e, y))^{1-d}. \quad (6.7)$$

For each lattice edge, send its signed F_y -value along its lifted simple path. To justify this infinite sum and estimate it, fix a target slot edge f and write $h = \text{dist}(f, y)$. If the route of an original edge e uses f , put $t = \text{dist}(e, y)$. The displacement bound gives

$$|h - t| \leq C(1 + t)^\nu + C.$$

Because $\nu < 1$, this implies $1 + t \asymp 1 + h$, with uniform constants: above a fixed threshold the error is at most $(1 + t)/2$, and below that threshold all quantities are bounded. It follows also that e lies within distance $C(1 + h)^\nu$ of f . There are at most

$$C(1 + h)^{d\nu} \quad (6.8)$$

such lattice edges. Each routed path is simple, so any one of them uses f at most once. Summing (6.7) over this finite set gives a well-defined flow J_a^0 with

$$|J_a^0(f)| \leq C(1 + h)^{1-d+d\nu}.$$

The count (6.8) already counts all contributions to f ; no path-length factor is present.

Local finiteness of the path sum and finiteness of the vertex preimages justify the divergence calculation at any slot b :

$$\begin{aligned} \text{div } J_a^0(b) &= \sum_{e=(u,v)} F_y(e) (\mathbf{1}_{\{\psi(u)=b\}} - \mathbf{1}_{\{\psi(v)=b\}}) \\ &= \sum_{z:\psi(z)=b} \text{div } F_y(z) = \mathbf{1}_{\{\psi(y)=b\}}. \end{aligned}$$

Here one fixed orientation is used for each original lattice edge; the signed route is always from $\psi(u)$ to $\psi(v)$. Adding the unit path flow from a to $\psi(y)$ therefore gives

$$J_a = J_a^0 + \mathbf{1}_{Q_a}, \quad \text{div } J_a = \mathbf{1}_{\{a\}}.$$

The connector has fixed bounded support near y , so enlarging C preserves the envelope (6.4). Finally, $2k > d$ for $d \geq 3$, and the number of slot edges at radius h is at most $C(1 + h)^{d-1}$. Squaring the envelope and summing proves finite energy. Equivalently, this is a unit flow to infinity: its net outward flux across any finite vertex set containing a is one. $\square$

6.3 Overlap and arbitrary subunit injections

We now combine the individual flows. Their decay, rather than a disjointness property of their routes, controls their total energy.

Lemma 6.3 (Overlap of two flows). *Suppose two flows on an available-slot graph obey*

$$|J(e)| \leq C_0(1 + \text{dist}(e, y))^{-k}, \quad |J'(e)| \leq C_0(1 + \text{dist}(e, y'))^{-k},$$

where $d/2 < k < d$. Then

$$\sum_e |J(e)J'(e)| \leq C(1 + |y - y'|_\infty)^{-(2k-d)}, \quad (6.9)$$

where C depends only on d, ℓ, k, C_0 .

Proof. Write $H = 1 + |y - y'|_\infty$. For bounded H the assertion follows by summing the squared envelopes and Cauchy–Schwarz. Otherwise, in the ball of radius $(H - 1)/3$ about either center, the other envelope is at most CH^{-k} , whereas the sum of the nearer envelope is at most CH^{d-k} . These two regions cost CH^{d-2k} . In the remaining region the two distances are comparable. Dyadic annuli of radii comparable to $2^j H$, $j \geq 0$, cost at most

$$C \sum_{j \geq 0} (2^j H)^{d-2k} \leq CH^{d-2k},$$

since $2k > d$. Changing a site distance to the distance from an incident edge changes it by at most one, and the number of slot edges per site is bounded by $2d\ell^2$. Thus the same estimates apply to the stated edge sum. $\square$

Proposition 6.4 (Energy for arbitrary injections). *Let $d \geq 3$, $\alpha = 1/100$, and $\nu = 1/(100d)$. Fix M_* as in (5.13) and take R sufficiently large as in Proposition 6.2. Let T be a finite set of m anchor slots with sites x_i , satisfying*

$$\#\{i \in T : x_i \in B_h(z)\} \leq L(1 + h)^\alpha \quad \text{for all } z \in \mathbb{Z}^d, \ h \geq 0.$$

Fix an available-slot graph and a subset $I \subset T$ of anchors regular for this graph. For each $i \in I$, let μ_i be any nonnegative mass on its slots, supported over $B_R(x_i)$ and satisfying $\sum_a \mu_i(a) \leq 1$. There is a flow J such that

$$\text{div } J = \sum_{i \in I} \mu_i, \quad \sum_e J(e)^2 \leq C|I|R^\alpha \leq CmR^\alpha. \quad (6.10)$$

Consequently every finitely supported real function g on the graph satisfies

$$\left| \left\langle \sum_{i \in I} \mu_i, g \right\rangle \right| \leq (CmR^\alpha \mathcal{D}(g))^{1/2}, \quad \mathcal{D}(g) = \frac{1}{2} \sum_e (g(e^-) - g(e^+))^2. \quad (6.11)$$

All constants are uniform over the choices of I and of the masses.

Proof. The empty case uses $J = 0$. Otherwise, for each slot a in the support of μ_i , take a unit flow $J_{i,a}$ from Proposition 6.2, using anchor x_i , and set

$$J = \sum_{i \in I} \sum_a \mu_i(a) J_{i,a}.$$

This is a finite linear combination, and its divergence is as asserted. Put $q = 2k - d = d - 2 - 2d\nu > \alpha$. By Lemma 6.3, the absolute overlap of $J_{i,a}$ and $J_{j,b}$ is at most $C(1 + |y_a - y_b|_\infty)^{-q}$, where y_a, y_b are the sites of their injection slots. If $|x_i - x_j|_\infty > 4R$, then $|y_a - y_b|_\infty \geq |x_i - x_j|_\infty/2$, uniformly over both supports. For the remaining anchor pairs use the uniform constant overlap bound. The mass bound of one per anchor therefore gives

$$\sum_e J(e)^2 \leq C \sum_{i \in I} \left[\#\{j \in I : |x_j - x_i|_\infty \leq 4R\} + \sum_{\substack{j \in I \\ |x_j - x_i|_\infty > 4R}} (1 + |x_j - x_i|_\infty)^{-q} \right].$$

The first term is at most CLR^α . Splitting the second into annuli $2^j 4R < |x_j - x_i|_\infty \leq 2^{j+1} 4R$ bounds it by

$$CL \sum_{j \geq 0} (2^j R)^{\alpha - q} \leq C,$$

because $R \geq 1$ and $q > \alpha$. This proves (6.10), including repeated anchor sites since the growth assumption counts slots.

Finally, finite support of g permits summation by divergence without a term at infinity:

$$\left\langle \sum_i \mu_i, g \right\rangle = \sum_e J(e)(g(e^-) - g(e^+)).$$

Cauchy-Schwarz and $\sum_e |\nabla_e g|^2 = 2\mathcal{D}(g)$ give (6.11). In particular it applies to functions that vanish on the killed boundary and outside the physical box, with exactly the energy normalization in (5.12). $\square$

The construction is deterministic at a single slice. Thus the masses and the set of regular anchors may depend on the whole exposed environment; the pairing bound requires neither independence nor a choice of flows continuous in time.

7 Heat-flow leakage and the pin bootstrap

We now use the geometric estimates to bound the return fraction in the simultaneous pin test. The main task is to prove that the hitting function loses nearly m units of squared norm during each of two short portions of a period, where m is the number of new pins. Local diffusion spreads the mass on those pins, and the flow estimate controls its pairing with the rest of the hitting function. Averaging the accumulated heat dissipation will absorb this pairing. The Nash method for Markov semigroups is developed in [10, Section 3]. The proof below accounts explicitly for the changing hole sets, killing, instantaneous contractions, and the restriction of the Nash inequality to regular times.

Throughout the section, $d \geq 3$, $\ell = 2S$, and Λ is a finite cube. We use the parameters of Section 5, namely

$$\alpha = \frac{1}{100}, \quad \nu = \frac{1}{100d}, \quad a = \frac{11}{10}, \quad s = p^{-a}, \quad R = \lceil C_* s \rceil, \quad \beta \geq 5s.$$

The spatial growth constant L , the regularity radius M_* , and the constant C_* will be fixed independently of the final choice of p . The test T consists of $m \geq 1$ distinct slot points at one time, is disjoint from $E \supseteq D_0$, and satisfies

$$\#\{i \in T : x_i \in B_h(z)\} \leq L(1 + h)^\alpha \quad (z \in \mathbb{Z}^d, h \geq 0). \quad (7.1)$$

Here x_i denotes the site of the slot point i . Write $P = E \cup T$, and sample the exposed paths under $\mathbb{P}_P$, as in Proposition 4.1. Expectations of quantities determined by these paths use their actual marginal, including the factor g_P in (4.4). At a fixed exposure, the propagators use the unconditioned fresh-turn walk. As in Proposition 3.1, r is the expected fraction of points of T whose next pin along the actual cycle again belongs to T .

Recall the one-period, column-forward kernel U of the fresh-turn walk, killed on visiting D_0 . At the cut containing T , the function f of Lemma 4.5 is 1 on T and, off T , is the probability of a future visit to T before killing, namely the minimal nonnegative hitting solution. In particular,

$$0 \leq f \leq 1, \quad f|_{\partial\Lambda} = 0, \quad U^*f = f \text{ off } T, \quad m(1-r) \geq \mathbb{E}_P\langle f, (I-U)f \rangle. \quad (7.2)$$

The last inequality incorporates the conditional first-moment identity of Proposition 4.3 and the relaxation of the other pins of E . A class of the relaxed walk that never reaches T has $f = 0$; certain absorption is unnecessary. In particular f is a function of the exposed paths, and we will not require it to be independent of any geometric event. All the propagators below are measurable functions of finite-jump data. The hitting function is measurable as the increasing limit of finite-horizon hitting probabilities, so the expectations used below are well defined.

For grid pin sets satisfying the larger-pin induction hypothesis $\mathfrak{l}_p(P)$ of (5.4), our aim is

$$\mathbb{E}_P\langle f, (I-U)f \rangle \geq m[1 - \varepsilon(p)], \quad \varepsilon(p) = o(p). \quad (7.3)$$

Together with (7.2), this gives $r \leq \varepsilon(p)$. The simultaneous pin test then recovers $\mathbb{P}_E(T \subseteq \mathcal{B}_E) \leq p^m$ once $2\varepsilon(p) \leq p$, closing the induction step. To obtain (7.3), split $f = \eta + g$, where $\eta = \mathbf{1}_T$ and $g = f - \eta$. Then $0 \leq g \leq 1$ and $\langle \eta, g \rangle = 0$. We first relate the quadratic form to norm losses at both ends of the period. The Nash estimate will make the propagated η small in squared norm, while the flow estimate will control its pairing with the propagated g through the square root of that function's energy at the same time.

7.1 Ordered propagation and the two ends of a period

We first fix one exposed environment. Between its finitely many instantaneous changes, the surviving interior holes carry the symmetric rate-1/2 heat equation. A boundary hole is assigned value zero. Thus the generator on an interior hole k is

$$(L_t v)_k = \frac{1}{2} \sum_{j: \{j,k\} \text{ is an available slot edge}} (\tilde{v}_j - v_k), \quad (7.4)$$

where $\tilde{v}$ equals v on interior holes and zero on boundary holes. The diagonal includes the escape rates to the boundary. This is continuous killing throughout the heat interval.

At an instantaneous change, let F be the forced bijection or the average of the allowed within-site completion bijections, from the hole set just before the change to that just after it. If R_- and R_+ are the projections deleting killed coordinates on those two sets, respectively, the killed factor is

$$R_+ F R_-, \quad (R_+ F R_-)^* = R_- F^* R_+. \quad (7.5)$$

The projections remain in this order. For heat intervals, the corresponding assertion follows either from the symmetric Dirichlet generator or by transposing the sampled surviving paths.

By the subpermutation argument in (4.14), these matrices, their products, and their adjoints are nonnegative, have row and column sums at most one, and contract the counting ℓ^2 norm. We

will also use the two sum bounds separately: the row bound gives $0 \leq Av \leq 1$ when $0 \leq v \leq 1$, while the column bound gives $\|Av\|_1 \leq \|v\|_1$ for $v \geq 0$.

Put the time of T at the beginning and end of a period. Let C_u be the initial propagation of duration u , and let C_u^{rev} be the adjoint of the ending propagation of duration u . Forward propagation includes the changes after its initial cut through its final cut, under the post-completion convention at time zero. Reverse propagation traverses precisely those factors in the opposite order. If the ending cut is a completion cut, its adjoint completion is the first factor traversed in reverse time. It is included exactly once, including when it occurs immediately at elapsed time zero. We set $C_0 = C_0^{\text{rev}} = I$ on the initial, already unkilld coordinates and count any such initial reverse jump in every positive duration that traverses it. This convention allows a contraction jump at $u = 0+$.

Lemma 7.1 (Leakage at both ends). *For every fixed exposed environment, $s > 0$, and period $\beta \geq 4s$, the preceding killed propagators satisfy, for every real vector v on the holes at the distinguished cut with zero boundary values,*

$$\langle v, (I - U)v \rangle \geq \frac{1}{2} \left(\|v\|_2^2 - \|C_{2s}v\|_2^2 + \|v\|_2^2 - \|C_{2s}^{\text{rev}}v\|_2^2 \right). \quad (7.6)$$

Both $u \mapsto \|C_u v\|_2$ and $u \mapsto \|C_u^{\text{rev}} v\|_2$ are nonincreasing.

Proof. The nonoverlapping initial and ending portions factor $U = BMA$, with $A = C_{2s}$, $B^* = C_{2s}^{\text{rev}}$, and M the intervening propagation. At $\beta = 4s$, assign every shared-cut jump once according to the preceding convention and take the intervening identity on that cut. All factors are contractions, so

$$\langle v, Uv \rangle = \langle B^*v, MAv \rangle \leq \|B^*v\|_2 \|Av\|_2 \leq \frac{1}{2} (\|B^*v\|_2^2 + \|Av\|_2^2).$$

This proves (7.6). Longer initial portions are obtained by left multiplication by further contraction factors. For the ending adjoints the same fact follows after reversing the factor order, using (7.5). It proves both monotonicity assertions. $\square$

The reverse construction has symmetric heat between changes, inverse forced transports, averages of inverse completion bijections, and the ordered killing projections. Thus it has the same deterministic analytic properties as forward propagation. Its environment is the original exposed environment read in reverse order; no invariance of the environment law under time reversal is asserted or needed.

Thus (7.3) will follow if each of these two propagations loses at least $m[1 - \varepsilon(p)]$ in expected squared norm by time $2s$. We prove both estimates through the same argument.

For the rest of the estimates, C_u denotes either of the two propagations, and $t(u)$ denotes its physical time slice modulo β . On the available slot graph, extended by all slots outside Λ , use the energy

$$\mathcal{D}_t(v) = \frac{1}{2} \sum_{\{j,k\} \text{ available at } t} (v_j - v_k)^2.$$

Vectors propagated in the box are extended by zero on its boundary and outside. Since the boundary is the inner boundary, any edge crossing out of the box has two zero values.

Write $\eta_u = C_u \eta$, $g_u = C_u g$, and $\mathcal{L}(u) = \|f\|_2^2 - \|C_u f\|_2^2$. For a jump traversed at elapsed time v , let

$$\Delta_g(v) = \|g_{v,\text{before}}\|_2^2 - \|g_{v,\text{after}}\|_2^2 \geq 0.$$

Here “before” and “after” refer to the chosen direction of propagation. Denote the finite list of jumps traversed by duration u by $\mathcal{J}(u)$, including an initial reverse jump when present. Between jumps, (7.4) gives

$$\frac{d}{du} \|g_u\|_2^2 = -2\mathcal{D}_{t(u)}(g_u).$$

To check the normalization, an interior–interior edge contributes $-(g_j - g_k)^2$ to this derivative, and an interior–boundary edge contributes $-g_j^2$. These are exactly the contributions to $-2\mathcal{D}$. Summing continuous and instantaneous losses and then expanding $\|\eta_u + g_u\|_2^2$ yields the exact identity

$$\begin{aligned} \mathcal{L}(u) &= m - \|\eta_u\|_2^2 - 2\langle \eta_u, g_u \rangle \\ &\quad + 2 \int_0^u \mathcal{D}_{t(v)}(g_v) dv + \sum_{v \in \mathcal{J}(u)} \Delta_g(v). \end{aligned} \quad (7.7)$$

The sum includes a completion at an endpoint whenever that completion is a factor of the propagation. Dropping it gives

$$\mathcal{L}(u) \geq m - \|\eta_u\|_2^2 - 2\langle \eta_u, g_u \rangle + 2 \int_0^u \mathcal{D}_{t(v)}(g_v) dv. \quad (7.8)$$

Keeping this as an inequality is necessary even when a completion is the only nontrivial instantaneous operation.

7.2 Localization with actual and fresh marks

For $i \in T$, let $\eta_{i,u}$ be the propagation from unit mass at i , with the additional rule that a path is killed when its site leaves $B_R(x_i)$. Use this additional killing for continuous moves and on both sides of instantaneous changes. Positivity and the path description give

$$0 \leq \zeta_u := \sum_{i \in T} \eta_{i,u} \leq \eta_u \leq 1, \quad \|\eta_{i,u}\|_1 \leq 1. \quad (7.9)$$

Lemma 7.2 (Localization averaged over the pin law). *Fix d and ℓ . There is a constant $C_* = C_*(d, \ell)$ such that for $s \geq 1$, $R = \lceil C_* s \rceil$, $\beta \geq 5s$, every finite cube, every $E \supseteq D_0$ consisting of D_0 and finitely many pins, and every disjoint single-time test T , one has*

$$\mathbb{E}_P \|\eta_u - \zeta_u\|_1 \leq C m e^{-cs} \quad (0 \leq u \leq 2s) \quad (7.10)$$

for either direction of propagation. The constants are independent of the volume, the pin sets, β , and s .

Proof. First sample a complete actual exchange picture with law $\mathbb{P}_P$ and expose its missing cycles. Conditional on that exposure, realize the unconditioned one-particle propagation of Proposition 4.1 using independent fresh rate-1/2 Poisson clocks on all slot edges, accepting a fresh mark only when its two endpoints are holes. The forced moves use marks of the actual picture. Completion randomness changes slots only within a site. This construction is for the unconditioned fresh walk; its exposure still has the marginal induced by the full pin law.

The difference in (7.9), summed over output slots, is bounded by the sum of the probabilities that these individual walks exit their respective balls before elapsed time u . Exiting requires at least R nearest-neighbor spatial steps. The first R such steps give a chronological list of marks,

each from either the actual picture or the fresh family. For a fixed list of sites, slot edges, and choices of source, the actual marks used are distinct: the time interval has length at most $2s < \beta$, so its image on the time circle does not use a mark twice. This remains true if the interval crosses time zero, and if it is traversed backwards. Fresh marks in the list are distinct as well.

Suppose a source choice uses k actual marks and $R - k$ fresh marks. By Lemma 2.4, the joint factorial measure of the actual marks is bounded by 2^k times their base product intensity. Independence of the fresh family then bounds the mixed measure by

$$1^k (1/2)^{R-k} dt_1 \cdots dt_R$$

on the prescribed edge coordinates. All source choices are integrated over the same ordered simplex $0 < t_1 < \cdots < t_R < u$, of volume $u^R/R!$. At each spatial step there are at most $2d\ell^2$ choices for the next site and the slot edge, even if one ignores constraints from the current slot. Summing source choices and edge choices therefore bounds the expected number of such lists, for one initial point, by

$$(2d\ell^2)^R (1 + 1/2)^R \frac{u^R}{R!} \leq \frac{(2Ks)^R}{R!}, \quad K = 3d\ell^2. \quad (7.11)$$

The existence of an exit path is bounded by this count; the counted lists need not themselves be realizable walk histories. In particular, the restrictions imposed by the exposure can be ignored when taking this upper bound. No factorial bound conditional on a fixed exposure has been used.

Since $R! \geq (R/e)^R$, choosing $C_* > 4eK$ bounds the last expression by $2^{-R} \leq e^{-Cs}$. Sum over $i \in T$ to obtain (7.10). Transposition merely reverses the chronological lists and the forced transports. The same mixed factorial estimate applies under the original law $\mathbb{P}_P$, which proves the reverse assertion too. $\square$

7.3 Diffusion during regular times

We now use the downward-induction hypothesis only through the three proved inputs from Sections 5 and 6. Assume $\mathbf{l}_p(P)$, as defined in (5.4). With the fixed choices of constants there, Proposition 5.4 gives

$$\mathbb{P}_P(t \text{ is not regular for } i) \leq \delta := CR^d p^{2d} + Ce^{-cR^\xi} \quad (i \in T, t \in [0, \beta)), \quad (7.12)$$

where $\xi > 0$ and the constants are fixed for all p below a preliminary threshold. At a regular slice, Proposition 5.6 states

$$\|v\|_2^{2+4/d} \leq C\mathcal{D}_t(v)\|v\|_1^{4/d} \quad (7.13)$$

for nonnegative v supported on surviving slots above $B_R(x_i)$, with zero extension. Finally, Proposition 6.4 applies to any subset of the anchors regular at that same time and to any nonnegative injections of mass at most one per anchor supported above its R -ball. Their sum is the divergence of a flow J with

$$\sum_e J(e)^2 \leq CmR^\alpha. \quad (7.14)$$

The flow estimate follows from Proposition 6.2 and the growth condition; its applicability to arbitrary injection points and subunit masses is part of its statement.

Lemma 7.3 (Decay after enough regular time). *Let T satisfy (7.1), and assume $\mathbf{l}_p(P)$ for $P = E \cup T$, with p below the preliminary threshold in Proposition 5.4. For the localized propagators*

defined above, in either direction and for $s \leq u \leq 2s$,

$$\mathbb{E}_P \|\eta_{i,u}\|_2^2 \leq C(s^{-d/2} + \delta) \quad (i \in T), \quad (7.15)$$

$$\mathbb{E}_P \|\eta_u\|_2^2 \leq Cm[e^{-cs} + R^\alpha(s^{-d/2} + \delta)]. \quad (7.16)$$

Proof. Put $q_i(u) = \|\eta_{i,u}\|_2^2$. All localized factors are contractions, so $0 \leq q_i \leq 1$ and q_i is nonincreasing, including at jumps. Between jumps the additional killing gives the Dirichlet generator with zero values outside $B_R(x_i)$ as well as on the physical boundary. Consequently, at every regular smooth time, (7.13) and $\|\eta_{i,u}\|_1 \leq 1$ imply

$$q'_i(u) = -2\mathcal{D}_{t(u)}(\eta_{i,u}) \leq -cq_i(u)^{1+2/d}. \quad (7.17)$$

At other smooth times $q'_i \leq 0$. As long as $q_i > 0$, the function $q_i^{-2/d}$ thus increases at a fixed positive rate during regular time and cannot decrease at other times or jumps. If q_i reaches zero, it stays zero. It follows in either case that

$$\int_0^s \mathbf{1}_{\{t(v) \text{ regular for } i\}} dv \geq s/2 \implies q_i(u) \leq Cs^{-d/2} \quad (s \leq u \leq 2s).$$

The expected amount of nonregular time in $[0, s]$ is at most $s\delta$ by (7.12) and Fubini. Markov's inequality bounds the probability of its exceeding $s/2$ by 2δ . This proves (7.15). For reverse propagation, $t(v)$ simply runs in the opposite direction; the same fixed-time estimate (7.12) gives the same integral bound, without changing the probability law.

At any slot above z , only anchors with $x_i \in B_R(z)$ can contribute to ζ_u . Their number is at most $L(1+R)^\alpha \leq CR^\alpha$ by (7.1). Hence pointwise Cauchy–Schwarz and summation give

$$\|\zeta_u\|_2^2 \leq CR^\alpha \sum_{i \in T} q_i(u).$$

Since $0 \leq \zeta_u \leq \eta_u \leq 1$,

$$\|\eta_u\|_2^2 - \|\zeta_u\|_2^2 = \sum_k (\eta_u(k) - \zeta_u(k))(\eta_u(k) + \zeta_u(k)) \leq 2\|\eta_u - \zeta_u\|_1.$$

Use Lemma 7.2 and (7.15) to obtain (7.16). $\square$

The initial pin mass has now spread sufficiently in mean square. The remaining term in (7.8) is its pairing with g_u . We next use flows at the very slice where this pairing occurs, so that the positive heat energy can absorb it.

7.4 Absorbing the interaction at the same time

Proposition 7.4 (Expected leakage). *Fix the constants in the preceding geometric estimates and choose C_* as in Lemma 7.2. For all sufficiently small $p > 0$, every finite cube, every circular grid of mesh at most p^{4d} , every E consisting of D_0 and grid pins, and every disjoint single-time grid test T satisfying (7.1), assume $\mathbf{l}_p(P)$ for $P = E \cup T$. If $\beta \geq 5s$, then, for each of the forward and reverse propagations,*

$$\mathbb{E}_P [\|f\|_2^2 - \|C_{2s}f\|_2^2] \geq m[1 - \varepsilon(p)], \quad (7.18)$$

where one may take

$$\varepsilon(p) = C \left[e^{-cs} + R^\alpha \left(s^{-d/2} + \delta + s^{-1} \right) \right], \quad \delta = CR^d p^{2d} + Ce^{-cR^\xi}. \quad (7.19)$$

All constants are independent of the volume, period, grid, pin sets, and final p .

Proof. For $s \leq u \leq 2s$, let

$$\mu_u = \sum_{i: t(u) \text{ regular for } i} \eta_{i,u}.$$

By (7.9), $0 \leq \mu_u \leq \eta_u$. The nonnegative discarded mass is bounded by the localization error plus the masses attached to nonregular anchors. Since $0 \leq g_u \leq 1$ and each local mass is at most one, Lemma 7.2 and (7.12) give

$$\mathbb{E}_P \langle \eta_u - \mu_u, g_u \rangle \leq Cm(e^{-cs} + \delta). \quad (7.20)$$

This uses only pointwise bounds, so dependence of g_u on the same environment causes no change.

At a fixed regular slice, each summand of μ_u meets exactly the support and mass hypotheses of (7.14). Choose a reference orientation for the available edges, with $\text{div } J$ equal to outgoing minus incoming flow. Since g_u is finitely supported, summation by parts has finitely many nonzero terms and gives

$$\langle \mu_u, g_u \rangle = \sum_{e=(j,k)} J(e)(g_u(j) - g_u(k)).$$

Cauchy-Schwarz, (7.14), and $\sum_e |\nabla_e g_u|^2 = 2\mathcal{D}_{t(u)}(g_u)$ imply the deterministic inequality

$$|\langle \mu_u, g_u \rangle| \leq (A\mathcal{D}_{t(u)}(g_u))^{1/2}, \quad A = CmR^\alpha. \quad (7.21)$$

It suffices that a flow exists at each time: the resulting inequality contains only measurable propagated vectors and energies, so no measurable or continuous choice of flows is required.

To make the remainder in weighted Young's inequality integrable at the final time, use an averaging density that vanishes linearly there; its tail weight then vanishes quadratically. Average (7.8) over $u \in [s, 2s]$ with probability density

$$w(u) = \frac{2(2s - u)}{s^2}, \quad W(u) = \int_u^{2s} w(v) dv = \frac{(2s - u)^2}{s^2}.$$

Since $\mathcal{L}$ is nondecreasing, $\mathcal{L}(2s) \geq \int_s^{2s} w(u)\mathcal{L}(u) du$. Tonelli's theorem shows that the positive energy in this average contains

$$2 \int_s^{2s} W(u) \mathcal{D}_{t(u)}(g_u) du;$$

the additional contribution from $[0, s]$ is nonnegative. At each $s \leq u < 2s$, weighted Young's inequality gives

$$2w(u)\sqrt{A\mathcal{D}_{t(u)}(g_u)} \leq 2W(u)\mathcal{D}_{t(u)}(g_u) + \frac{Aw(u)^2}{2W(u)}. \quad (7.22)$$

The same-time energy has therefore absorbed the retained pairing. The remaining integral is finite, because

$$\int_s^{2s} \frac{w(u)^2}{W(u)} du = \frac{4}{s}. \quad (7.23)$$

The endpoint $u = 2s$ has measure zero; the quotient has the constant value $4/s^2$ on $[s, 2s)$.

Taking expectation, using (7.20), and then Lemma 7.3, we conclude that

$$\begin{aligned} \mathbb{E}_P \mathcal{L}(2s) &\geq m - Cm[e^{-cs} + R^\alpha(s^{-d/2} + \delta)] - Cm(e^{-cs} + \delta) - CmR^\alpha/s \\ &\geq m \left[1 - C \left(e^{-cs} + R^\alpha(s^{-d/2} + \delta + s^{-1}) \right) \right]. \end{aligned}$$

Here $R \geq 1$ allows the unweighted δ error to be included in $R^\alpha\delta$. Every step applied to either ordered propagation, which proves both assertions. $\square$

7.5 Choosing the constants and closing the finite induction

The error has enough room to recover the assumed sparsity with the same parameter p . Indeed, after C_* is fixed, R is comparable to $s = p^{-a}$ for $p \leq 1$. The nonexponential errors R^α/s , $R^\alpha s^{-d/2}$, and $R^{\alpha+d} p^{2d}$ in (7.19) are bounded by constant multiples of

$$p^{a(1-\alpha)}, \quad p^{a(d/2-\alpha)}, \quad p^{2d-a(d+\alpha)}, \quad (7.24)$$

respectively. Their exponents satisfy

$$a(1-\alpha) = \frac{1089}{1000} > 1, \quad a(d/2-\alpha) \geq \frac{1639}{1000} > 1, \quad 2d-a(d+\alpha) \geq \frac{2689}{1000} > 1.$$

The last two lower bounds use $d \geq 3$; each exponent increases with d . Both e^{-cs} and $R^\alpha e^{-cR^\xi}$ are smaller than every fixed power of p . Thus

$$\varepsilon(p) = o(p) \quad (p \downarrow 0). \quad (7.25)$$

For clarity, the constant choices can be made in the following order. Fix d, ℓ, α, ν, a . Fix the witness scale b and then its growth constant $L \geq \max\{2d, \ell, 4\}$ as in Lemma 5.2. Fix the local connectivity radii and a sufficiently large M_* for Proposition 5.6 and the hull and flow constructions of Lemma 6.1 and Proposition 6.2. Fix a preliminary $p_0 > 0$ that absorbs the admissible witness count in Lemma 5.2; retain the resulting fixed constants in Proposition 5.4, including c and ξ . Next choose C_* for Lemma 7.2. Finally reduce $p \leq p_0$ so that R exceeds the fixed local radii and the fixed lower bounds needed for the flow comparisons, $p < 1/4$, and $2\varepsilon(p) \leq p$. The limit (7.25) permits all of these final requirements simultaneously. No constant in the component tail is recomputed using this final p .

Theorem 7.5 (Uniform pin estimate). *For every integer $d \geq 3$ and integer slot multiplicity $\ell \geq 1$, there exist $L \geq \max\{2d, \ell, 4\}$ and $p_* \in (0, 1/4)$ with the following property. Let $0 < p \leq p_*$ and $\beta \geq 5p^{-11/10}$. In any finite cube Λ , choose a finite circular time grid containing zero and of mesh at most p^{4d} . Let E consist of D_0 and any subset of the slot points on this grid. Then every nonempty test T at a single grid time, disjoint from E and satisfying (7.1), obeys*

$$\mathbb{P}_E(T \subseteq \mathcal{B}_E) \leq p^{|T|}. \quad (7.26)$$

In particular, with $p = p_$ and $\beta_0 = 5p_*^{-11/10} < \infty$, for every finite $\beta \geq \beta_0$, every cube, and every slot j at time zero,*

$$\mathbb{P}_{D_0}(j \in \mathcal{B}_{D_0}) \leq p_*. \quad (7.27)$$

The constants depend only on d and ℓ .

Proof. Fix all auxiliary constants in the stated order and choose p_* so that every $0 < p \leq p_*$ meets the final requirements. The set of interior slot points on a fixed finite grid is finite. Induct on the number of these points missing from E . When none is missing there is no disjoint nonempty test, so $\mathbf{l}_p(E)$ holds vacuously.

Assume the assertion for pin sets with fewer missing grid points than E , and let T be a test for E . Since T is nonempty and disjoint from E , the set $P = E \cup T$ has strictly fewer missing points. The induction hypothesis therefore supplies $\mathbf{l}_p(P)$, exactly the hypothesis used in Proposition 5.4 and Proposition 7.4. The two leakage estimates and Lemma 7.1 give

$$\mathbb{E}_P\langle f, (I - U)f \rangle \geq m(1 - \varepsilon(p)).$$

By Lemma 4.5, or (7.2), it follows that $r \leq \varepsilon(p)$. Proposition 3.1 now yields

$$\mathbb{P}_E(T \subseteq \mathcal{B}_E) \leq (2r)^m \leq (2\varepsilon(p))^m \leq p^m.$$

This proves the induction step for every allowed T , and finiteness of the grid completes the induction. A singleton interior slot test at time zero satisfies the growth bound, so taking $E = D_0$ gives (7.27). A boundary slot belongs to D_0 , and its missing-cycle probability is zero. This covers all slots. $\square$

Theorem 7.5 supplies the uniform missing-cycle bound required for the magnetization argument. The grid and the larger pin sets were devices for a finite induction; the conclusion (7.27) concerns the original continuous boundary pins alone.

8 From pinned loops to a magnetized equilibrium state

Theorem 7.5 says that, under boundary pinning, each slot has small probability of belonging to a cycle that misses the boundary. We now turn that statement into a state for the original zero-field dynamics. First we obtain a tail bound for the exact spectral distribution of total magnetization in a free box. A field tending to zero selects that tail, and translation averages produce a state with positive magnetization. The last step verifies both boundary values of the KMS strip for that same state.

Fix $d \geq 3$ and $S \in \{\frac{1}{2}, 1, \frac{3}{2}, \dots\}$. Let p_* be the threshold supplied by Theorem 7.5 for $\ell = 2S$, and put $p = p_*$ and $\beta_0(d, S) = 5p_*^{-11/10}$. Throughout this Section, $\beta \geq \beta_0(d, S)$ is fixed and finite. Write

$$\Lambda_N = [-N, N]^d \cap \mathbb{Z}^d, \quad V_N = |\Lambda_N|, \quad M_N = \sum_{x \in \Lambda_N} S_x^z, \quad H_N = H_{\Lambda_N}.$$

The quantity V_N here is the number of sites; the slot set remains $V_{\Lambda_N} = \Lambda_N \times \{1, \dots, \ell\}$, with $\ell = 2S$. Let μ_N be the spectral distribution of M_N in the free zero-field Gibbs state:

$$\mu_N(I) = \frac{\text{Tr}(e^{-\beta H_N} \mathbf{1}_I(M_N))}{\text{Tr}(e^{-\beta H_N})} \quad \text{for Borel sets } I \subset \mathbb{R}.$$

By Proposition 2.2, this is exactly the time-zero magnetization distribution in the cycle-coloring representation.

8.1 A spectral tail with surface-order cost

Lemma 8.1 (Boundary coloring and a spectral tail). *For every $N \geq 1$, let G_N be the event in the zero-field mark-and-color law that every cycle meeting D_0 is colored up. There is a constant $C_{\beta,d,S} < \infty$, independent of N , such that*

$$\Pr(G_N) \geq \exp\{-C_{\beta,d,S}|\partial\Lambda_N|\}, \tag{8.1}$$

$$\mu_N([SV_N/2, \infty)) \geq \frac{1}{2} \exp\{-C_{\beta,d,S}|\partial\Lambda_N|\}. \tag{8.2}$$

Conditionally on G_N , the uncolored picture has law $\mathbb{P}_{D_0}$, and the cycles missing D_0 still have independent fair colors.

Proof. Let n_0 be the number of cycles meeting D_0 , and let J_N count inter-site marks incident to a boundary site during one period. Every cycle meeting the boundary either contains a time-zero boundary slot or uses an inter-site mark incident to the boundary. Indeed, in the absence of a time-zero boundary slot on that cycle, a boundary visit must begin by a spatial jump; within-site endpoint permutations do not change the site. Each mark has two strands and can witness at most two cycles. Consequently, picture by picture,

$$n_0 \leq \ell |\partial \Lambda_N| + 2J_N.$$

There are at most $2d\ell^2 |\partial \Lambda_N|$ slot edges incident to the boundary. Their base rate is $1/2$. The first factorial bound of Lemma 2.4, applied with no pins, therefore gives

$$\mathbb{E}_\emptyset J_N \leq 2d\beta\ell^2 |\partial \Lambda_N|, \quad \mathbb{E}_\emptyset n_0 \leq (\ell + 4d\beta\ell^2) |\partial \Lambda_N|.$$

Given a picture, G_N has probability 2^{-n_0} . Jensen's inequality now proves (8.1), for example with $C_{\beta,d,S} = (\log 2)(\ell + 4d\beta\ell^2)$.

Multiplying the zero-field picture weight $2^{\#\text{cycles}}$ by 2^{-n_0} leaves precisely the weight for cycles missing D_0 . This proves the assertion about the conditional picture; conditioning fixes only the colors of the other cycles. For a time-zero slot i , its conditional mean spin is thus

$$\frac{1}{2}(1 - \mathbb{P}_{D_0}(i \in \mathcal{B}_{D_0})).$$

For an unpinned slot the probability on the right is at most p by the singleton case of Theorem 7.5; for a boundary slot it is zero. Summing over the ℓV_N slots gives

$$\mathbb{E} \left[\frac{M_N}{V_N} \middle| G_N \right] \geq S(1 - p).$$

Here M_N denotes the random variable in the coloring representation. If $q = \Pr(M_N/V_N \geq S/2 \mid G_N)$, its upper bound $M_N/V_N \leq S$ implies

$$S(1 - p) \leq Sq + \frac{S}{2}(1 - q), \quad q \geq 1 - 2p > \frac{1}{2}.$$

Multiply by $\Pr(G_N)$ and use the exact spectral interpretation of the unconditioned coloring law to obtain (8.2). $\square$

The boundary event has not changed the Hamiltonian whose spectral law appears in (8.2). It supplies a tail event in the free Gibbs state whose logarithmic cost is of order the boundary size. We next select this tail by a homogeneous field whose volume contribution is larger than that cost, although the field itself tends to zero. The relation between the zero-field magnetization distribution and field-selected spontaneous magnetization goes back to Griffiths [16]; see also [14, Section 1]. Here the surface-order tail permits the explicit volume-dependent field used below.

Proposition 8.2 (An explicit vanishing-field limit). *For $N \geq 1$, set*

$$h_N = N^{-1/2}, \quad K_N = H_N - h_N M_N, \quad \rho_N^{\text{in}}(C) = \frac{\text{Tr}(e^{-\beta K_N} C)}{\text{Tr}(e^{-\beta K_N})} \quad (C \in \mathcal{A}_{\Lambda_N}).$$

Extend ρ_N^{in} to a state ρ_N on $\mathcal{A}$ by the product normalized trace outside Λ_N , and define

$$\omega_N(A) = \frac{1}{V_N} \sum_{x \in \Lambda_N} \rho_N(T_x(A)). \quad (8.3)$$

There is a subsequence $N_j \rightarrow \infty$ on which ω_{N_j} converges pointwise on $\mathcal{A}$ to a translation-invariant state ω , and every such subsequential limit satisfies

$$\omega(S_0^z) \geq S/4. \quad (8.4)$$

Proof. Each interaction term commutes with the sum of the two endpoint z -spins. This follows directly from

$$\mathbf{S}_x \cdot \mathbf{S}_y = S_x^z S_y^z + \frac{1}{2}(S_x^+ S_y^- + S_x^- S_y^+).$$

Thus $[H_N, M_N] = 0$. For any $h > 0$, the spectral law in the Gibbs state of $H_N - hM_N$ is exactly

$$\mu_{N,h}(\mathrm{d}m) = \frac{e^{\beta h m} \mu_N(\mathrm{d}m)}{\int e^{\beta h u} \mu_N(\mathrm{d}u)}.$$

Bound the numerator on $m < SV_N/4$, and retain only $m \geq SV_N/2$ in the denominator. Lemma 8.1 then gives

$$\mu_{N,h}((-\infty, SV_N/4)) \leq 2 \exp \left\{ C_{\beta,d,S} |\partial \Lambda_N| - \frac{\beta h S}{4} V_N \right\}. \quad (8.5)$$

For these cubes,

$$\frac{|\partial \Lambda_N|}{V_N} = 1 - \left(\frac{2N-1}{2N+1} \right)^d \leq \frac{2d}{2N+1}.$$

At $h = h_N$, the exponent in (8.5) tends to $-\infty$: its negative term has order $N^{d-1/2}$, whereas its positive term has order N^{d-1} . Hence, writing $q_N = \mu_{N,h_N}((-\infty, SV_N/4))$, we have $q_N \rightarrow 0$. The spectral bounds $-S \leq M_N/V_N \leq S$ imply

$$\rho_N^{\text{in}}(M_N/V_N) \geq \frac{S}{4}(1 - q_N) - S q_N = \frac{S}{4} - \frac{5S}{4} q_N. \quad (8.6)$$

In particular this estimate uses a single explicit volume-field sequence for the fixed β .

Choose a countable norm-dense set in the union of local matrix algebras, closed under rational complex linear combinations and adjoints. The bound $|\omega_N(A)| \leq \|A\|$ permits diagonal subsequence selection on that set. Norm approximation then gives pointwise convergence on $\mathcal{A}$. The resulting functional ω is linear, normalized and positive, since these properties hold for each ω_N and pass to pointwise limits. It is therefore a state.

For every fixed $y \in \mathbb{Z}^d$ and $A \in \mathcal{A}$,

$$|\omega_N(T_y(A)) - \omega_N(A)| \leq \frac{|(\Lambda_N + y) \triangle \Lambda_N|}{V_N} \|A\| \rightarrow 0.$$

This proves translation invariance of ω . Finally,

$$\omega_N(S_0^z) = \frac{1}{V_N} \sum_{x \in \Lambda_N} \rho_N(S_x^z) = \rho_N^{\text{in}}(M_N/V_N).$$

Taking the limit in (8.6) proves (8.4). □

8.2 Dynamics far from the translated boundary

Fix from now on one state subsequence N_j provided by Proposition 8.2. We will prove the KMS condition along this same subsequence for every local pair; no further subsequence depending on that pair will be taken. The needed dynamical approximation must hold uniformly over translations whose distance from the box boundary diverges.

We first check the hypotheses of the dynamics existence theorem cited in Section 1.1, [30, Theorem 2.2]. Use the lattice graph metric and the summable function $F(r) = (1+r)^{-d-1}$, which has the convolution bound required there. The interaction is supported on nearest-neighbor pairs, each term has norm at most $3S^2$, and every site belongs to $2d$ such pairs. Thus its interaction norm in that theorem is finite, with parameter $a = 0$.

For a finite cube Γ , write $\tau_t^\Gamma(B) = e^{itH_\Gamma} B e^{-itH_\Gamma}$, acting trivially outside Γ . Likewise let

$$\alpha_z^N(B) = e^{izK_N} B e^{-izK_N} \quad (z \in \mathbb{C}).$$

For real t , these maps are isometric automorphisms. For complex z , α_z^N is defined on the entire quasi-local algebra because K_N is a bounded local operator.

Lemma 8.3 (Uniform approximation in deep translated boxes). *Let B be a local observable, let L be a finite set containing $\text{supp } B$ and the origin, and let $0 < T < \infty$. Then*

$$\lim_{r \rightarrow \infty} \sup_{\substack{\Gamma \text{ a finite cube} \\ \text{dist}_\infty(L, \Gamma^c) \geq r}} \sup_{|t| \leq T} \|\tau_t^\Gamma(B) - \tau_t(B)\| = 0. \quad (8.7)$$

Here $\Gamma^c = \mathbb{Z}^d \setminus \Gamma$. If

$$I_N = \{x \in \Lambda_N : \text{dist}_\infty(x, \Lambda_N^c) \geq \lfloor \sqrt{N} \rfloor\},$$

then $|\Lambda_N \setminus I_N|/V_N \rightarrow 0$ and

$$\lim_{N \rightarrow \infty} \sup_{x \in I_N} \sup_{|t| \leq T} \|\alpha_t^N(T_x(B)) - T_x(\tau_t(B))\| = 0. \quad (8.8)$$

Proof. First fix t . If (8.7) failed at that time, there would be $\varepsilon > 0$ and cubes Γ_k whose distances from L to their complements tend to infinity, with $\|\tau_t^{\Gamma_k}(B) - \tau_t(B)\| \geq \varepsilon$. Such cubes contain centered boxes with radii tending to infinity. We can therefore select a subsequence that is increasing by inclusion and exhausts $\mathbb{Z}^d$: after any finite selected cube, a sufficiently late cube contains it and the next prescribed centered box. This contradicts the norm convergence along increasing boxes in the definition of τ_t .

To make this convergence uniform in time, write $\Phi(\{u, v\}) = -\mathbf{S}_u \cdot \mathbf{S}_v$. Only edges meeting L contribute to $[H_\Gamma, B]$, so for every Γ

$$\|[H_\Gamma, B]\| \leq 2\|B\| \sum_{\substack{e \text{ nearest-neighbor edge} \\ e \cap L \neq \emptyset}} \|\Phi(e)\| =: c_B < \infty.$$

Consequently $\|\tau_t^\Gamma(B) - \tau_u^\Gamma(B)\| \leq c_B|t - u|$. Passing to the infinite-volume limit gives the same bound for $\tau_t(B)$. A finite mesh of $[-T, T]$, together with the fixed-time conclusion, proves (8.7).

For $x \in I_N$, set $\Gamma = \Lambda_N - x$. Then $\text{dist}_\infty(L, \Gamma^c) \rightarrow \infty$ uniformly over these x , since L is fixed. For all sufficiently large N , $L \subset \Gamma$. Define the local rotation

$$\gamma_\theta^L(B) = e^{i\theta \sum_{u \in L} S_u^z} B e^{-i\theta \sum_{u \in L} S_u^z}.$$

The commutation $[H_N, M_N] = 0$ and translation of the finite Hamiltonian give the exact formula

$$T_{-x}(\alpha_t^N(T_x(B))) = \tau_t^\Gamma(\gamma_{-h_N t}^L(B)). \quad (8.9)$$

The norm effect of this rotation is bounded by

$$\|\gamma_{-h_N t}^L(B) - B\| \leq h_N |t| \left\| \left[\sum_{u \in L} S_u^z, B \right] \right\|.$$

Use the isometry of τ_t^Γ , followed by (8.7), to prove (8.8). Finally, the excluded sites lie in a layer of thickness at most $\lfloor \sqrt{N} \rfloor$, whose fraction of the cube is $O(N^{-1/2})$. $\square$

8.3 The analytic strip and its two boundary values

We will use the following elementary strip lemma. Its first part establishes a common strip bound from finite, possibly nonuniform interior bounds. Only after obtaining that common bound does its second part use Gaussian decay to control convergence on the whole boundary.

Lemma 8.4 (Bounded strip limits). *Fix $\beta > 0$, and put $\mathcal{S}_\beta = \{z \in \mathbb{C} : 0 \leq \text{Im } z \leq \beta\}$. Let $(F_n)_{n \geq 1}$ be functions continuous on $\mathcal{S}_\beta$, analytic in its interior, and individually bounded there. Suppose a constant $C \geq 0$ satisfies*

$$|F_n(t)| \leq C, \quad |F_n(t + i\beta)| \leq C \quad (n \geq 1, t \in \mathbb{R}).$$

Then $|F_n(z)| \leq C$ throughout $\mathcal{S}_\beta$. If the two boundary sequences converge locally uniformly on $\mathbb{R}$ to f_0 and f_β , respectively, then the full sequence F_n converges locally uniformly on the closed strip to a bounded function F , continuous on the closed strip and analytic in its interior, with

$$|F(z)| \leq C, \quad F(t) = f_0(t), \quad F(t + i\beta) = f_\beta(t).$$

Proof. Fix n , and let $M_n < \infty$ bound $|F_n|$ on the strip. For $\varepsilon > 0$, apply the maximum principle to $e^{-\varepsilon z^2} F_n(z)$ on $[-R, R] + i[0, \beta]$. Its modulus is at most $C e^{\varepsilon \beta^2}$ on the horizontal sides and at most $M_n e^{-\varepsilon R^2 + \varepsilon \beta^2}$ on the vertical sides. Send $R \rightarrow \infty$ while holding a point z fixed, then send $\varepsilon \downarrow 0$. This proves $|F_n(z)| \leq C$. Only the already assumed finite bound M_n was needed to remove the vertical sides.

Now set $G_n(z) = e^{-z^2} F_n(z)$. The common bound just proved implies

$$|G_n(t + iy)| \leq C e^{-t^2 + y^2} \quad (0 \leq y \leq \beta).$$

Local uniform convergence of the two boundary sequences and the Gaussian tail therefore make (G_n) uniformly Cauchy on both horizontal edges. Write

$$\delta_{n,m} = \max_{y \in \{0, \beta\}} \sup_{t \in \mathbb{R}} |G_n(t + iy) - G_m(t + iy)|; \quad \delta_{n,m} \longrightarrow 0.$$

The maximum principle on the same rectangles, now applied to $G_n - G_m$, has horizontal bound $\delta_{n,m}$ and vertical bound $2C e^{-R^2 + \beta^2}$. Letting $R \rightarrow \infty$ shows

$$\sup_{z \in \mathcal{S}_\beta} |G_n(z) - G_m(z)| \leq \delta_{n,m}.$$

Hence G_n converges uniformly on the closed strip to a continuous function G , analytic in the interior. Multiplication by e^{z^2} gives local uniform convergence of F_n on the closed strip to $F = e^{z^2} G$. Its claimed boundary values and the bound $|F| \leq C$ follow by taking pointwise limits. $\square$

Completion of the proof of Theorem 1.1. Let ω and N_j be as in Proposition 8.2. For local observables A, B , define the entire functions

$$F_N^{A,B}(z) = \frac{1}{V_N} \sum_{x \in \Lambda_N} \rho_N(T_x(A) \alpha_z^N(T_x(B))). \quad (8.10)$$

Trace cyclicity gives both identities

$$F_N^{A,B}(t) = \frac{1}{V_N} \sum_{x \in \Lambda_N} \rho_N(T_x(A) \alpha_t^N(T_x(B))), \quad (8.11)$$

$$F_N^{A,B}(t + i\beta) = \frac{1}{V_N} \sum_{x \in \Lambda_N} \rho_N(\alpha_t^N(T_x(B)) T_x(A)). \quad (8.12)$$

These formulas also hold when the translated supports cross the boundary of Λ_N . To see this without a support restriction, take a finite set containing Λ_N and both translated supports. On its matrix algebra the density of ρ_N is $e^{-\beta K_N} / \text{Tr}(e^{-\beta K_N})$ tensored with a normalized identity. Equivalently, it is the Gibbs density for the operator $K_N \otimes I$ on that larger algebra. The dynamics is generated by that same operator, so finite-dimensional trace cyclicity proves (8.12) there as well.

For real t , the isometry of α_t^N bounds both right-hand sides by $\|A\| \|B\|$. Before using any common strip bound, we also have the finite bound

$$|F_N^{A,B}(t + iy)| \leq \|A\| \|B\| e^{2\beta \|K_N\|} \quad (t \in \mathbb{R}, 0 \leq y \leq \beta).$$

Indeed, the real-time unitary factors have norm one, and $\|e^{\pm y K_N}\| \leq e^{\beta \|K_N\|}$. The first part of Lemma 8.4 therefore applies and gives the uniform bound

$$|F_N^{A,B}(z)| \leq \|A\| \|B\| \quad (z \in \mathcal{S}_\beta, N \geq 1). \quad (8.13)$$

We next identify the two boundary limits. Given $0 < T < \infty$, Lemma 8.3 supplies numbers $\varepsilon_N(T) \rightarrow 0$ bounding the norm error in (8.8) on I_N . On its complement the error is at most $2\|B\|$, since both real-time dynamics are isometric. It follows that, uniformly for $|t| \leq T$,

$$\left| F_N^{A,B}(t) - \omega_N(A \tau_t(B)) \right| \leq \|A\| \varepsilon_N(T) + 2\|A\| \|B\| \frac{|\Lambda_N \setminus I_N|}{V_N}, \quad (8.14)$$

$$\left| F_N^{A,B}(t + i\beta) - \omega_N(\tau_t(B) A) \right| \leq \|A\| \varepsilon_N(T) + 2\|A\| \|B\| \frac{|\Lambda_N \setminus I_N|}{V_N}. \quad (8.15)$$

For example, the comparison term in the first line is exactly $V_N^{-1} \sum_x \rho_N(T_x(A) T_x(\tau_t(B)))$, which equals $\omega_N(A \tau_t(B))$; the second line retains the opposite operator order. Thus both error bounds tend to zero.

The maps $t \mapsto A \tau_t(B)$ and $t \mapsto \tau_t(B) A$ are norm-continuous. Their images of a compact time interval are norm-compact. Pointwise convergence $\omega_{N_j}(C) \rightarrow \omega(C)$ is uniform over each such image: use a finite norm net and the common functional norm one. Therefore, along the already chosen subsequence,

$$\begin{aligned} F_{N_j}^{A,B}(t) &\longrightarrow \omega(A \tau_t(B)), \\ F_{N_j}^{A,B}(t + i\beta) &\longrightarrow \omega(\tau_t(B) A) \end{aligned}$$

locally uniformly for real t . Lemma 8.4 now gives a bounded continuous function $F_{A,B}$ on the closed strip, analytic in its interior, with precisely these two boundary values and

$$\sup_{z \in \mathcal{S}_\beta} |F_{A,B}(z)| \leq \|A\| \|B\|. \quad (8.16)$$

Since the strip limit uses the full state subsequence, the bilinearity of (8.10) passes to $F_{A,B}$.

It remains to obtain the asserted condition on the entire algebra. Let A_n, B_n be local observables converging in norm to arbitrary $A, B \in \mathcal{A}$. Bilinearity and (8.16) imply

$$\sup_{z \in \mathcal{S}_\beta} |F_{A_n, B_n}(z) - F_{A_m, B_m}(z)| \leq \|A_n - A_m\| \|B_n\| + \|A_m\| \|B_n - B_m\|.$$

Hence these functions converge uniformly on the closed strip to a bounded continuous $F_{A,B}$, analytic in the interior. The same inequality proves independence of the approximating sequences and the bound $\sup |F_{A,B}| \leq \|A\| \|B\|$. Since states have norm one and τ_t is isometric, norm approximation passes both boundary identities to the limit, uniformly in real t :

$$F_{A,B}(t) = \omega(A\tau_t(B)), \quad F_{A,B}(t + i\beta) = \omega(\tau_t(B)A).$$

This is the full β -KMS condition (1.2) for the zero-field dynamics τ defined in Section 1.1. The state is translation invariant and satisfies $\omega(S_0^z) \geq S/4 > 0$ by Proposition 8.2. Since β was any fixed finite number at least $\beta_0(d, S)$, Theorem 1.1 follows. $\square$

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